Supplementary Material — Duplicate Data Rows Make Prediction Uncertainty Look Better Than It Is: A Simple Check and Fix for New Cases

All tables are generated by `supplementary/build_supplement.py` from the result files in `results/`. Coverage gaps are in proportions (multiply by 100 for percentage points).

S1 Dataset audit

One row per dataset (OpenML data ID). `n_raw`: rows before group-preserving subsampling to 30,000. `dup_row_frac`: share of rows in a group of identical feature vectors of size ≥ 2. `conflict_row_frac`: share of rows in groups with more than one label. `pred_exposure`: Proposition 2. `obs_exposure`: mean over 10 random splits. `set`: DUP ($\pi' \geq 0.01$), LOW ($0 < \pi' < 0.01$), CLEAN (no duplicates).

task	data_id	name	n_rows	n_raw	n_features	n_classes	dup_row_frac	conflict_row_frac	max_group	pred_exposure	obs_exposure	set
classification	469	analcatdata_dmft	797	797	4	6.000	0.893	0.862	29	0.811	0.810	DUP
classification	4534	PhishingWebsites	11055	11055	30	2.000	0.713	0.032	25	0.576	0.572	DUP
classification	1067	kc1	2109	2109	21	2.000	0.519	0.120	112	0.475	0.478	DUP
classification	15	breast-w	699	699	9	2.000	0.406	0.000	27	0.377	0.379	DUP
classification	1464	blood-transfusion-service-center	748	748	4	2.000	0.421	0.267	35	0.367	0.374	DUP
classification	1063	kc2	522	522	21	2.000	0.356	0.226	63	0.320	0.299	DUP
classification	40978	Internet-Advertisements	3279	3279	1558	2.000	0.347	0.016	58	0.300	0.291	DUP
classification	1053	jm1	10885	10885	21	2.000	0.251	0.082	173	0.214	0.215	DUP
classification	1068	pc1	1109	1109	21	2.000	0.226	0.012	31	0.171	0.167	DUP
classification	40984	segment	2310	2310	18	7.000	0.203	0.002	6	0.126	0.122	DUP
classification	1475	first-order-theorem-proving	6118	6118	51	6.000	0.173	0.055	21	0.121	0.123	DUP
classification	44	spambase	4601	4601	57	2.000	0.125	0.001	69	0.099	0.096	DUP
classification	23	cmc	1473	1473	9	3.000	0.143	0.095	4	0.095	0.089	DUP
classification	1050	pc3	1563	1563	37	2.000	0.123	0.006	35	0.094	0.104	DUP
classification	1049	pc4	1458	1458	37	2.000	0.115	0.002	31	0.091	0.096	DUP
classification	6	letter	20000	20000	16	26.000	0.109	0.000	26	0.081	0.078	DUP
classification	1486	nomao	30000	34465	118	2.000	0.106	0.000	136	0.079	0.079	DUP
classification	40670	dna	3186	3186	180	3.000	0.097	0.001	5	0.071	0.069	DUP
classification	46	splice	3190	3190	60	3.000	0.097	0.001	5	0.071	0.072	DUP
classification	18	mfeat-morphological	2000	2000	6	10.000	0.111	0.107	4	0.067	0.066	DUP
classification	1468	cnae-9	1080	1080	856	9.000	0.047	0.000	4	0.032	0.035	DUP
classification	1480	ilpd	583	583	10	2.000	0.045	0.000	2	0.027	0.024	DUP
classification	1462	banknote-authentication	1372	1372	4	2.000	0.026	0.000	4	0.022	0.019	DUP
classification	38	sick	3772	3772	29	2.000	0.025	0.000	6	0.020	0.021	DUP
classification	40499	texture	5500	5500	40	11.000	0.010	0.000	2	0.006	0.007	LOW
classification	40983	wilt	4839	4839	5	2.000	0.008	0.000	2	0.005	0.004	LOW
classification	14	mfeat-fourier	2000	2000	76	10.000	0.006	0.000	2	0.004	0.003	LOW
classification	12	mfeat-factors	2000	2000	216	10.000	0.006	0.000	2	0.004	0.003	LOW
classification	22	mfeat-zernike	2000	2000	47	10.000	0.006	0.000	2	0.004	0.003	LOW
classification	40979	mfeat-pixel	2000	2000	240	10.000	0.006	0.000	2	0.004	0.003	LOW
classification	16	mfeat-karhunen	2000	2000	64	10.000	0.006	0.000	2	0.004	0.003	LOW
classification	1487	ozone-level-8hr	2534	2534	72	2.000	0.004	0.000	9	0.004	0.003	LOW
classification	1494	qsar-biodeg	1055	1055	41	2.000	0.006	0.000	2	0.003	0.001	LOW
classification	23381	dresses-sales	500	500	12	2.000	0.004	0.004	2	0.002	0.004	LOW
classification	1489	phoneme	5404	5404	5	2.000	0.003	0.000	2	0.002	0.002	LOW
classification	1590	adult	30000	48842	14	2.000	0.002	0.000	2	0.001	0.001	LOW
classification	4538	GesturePhaseSegmentationProcessed	9873	9873	32	5.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	188	eucalyptus	736	736	19	5.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	40994	climate-model-simulation-crashes	540	540	18	2.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	151	electricity	30000	45312	8	2.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	458	analcatdata_authorship	841	841	70	4.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	37	diabetes	768	768	8	2.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	54	vehicle	846	846	18	4.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	182	satimage	6430	6430	36	6.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	50	tic-tac-toe	958	958	9	2.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	1510	wdbc	569	569	30	2.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	3	kr-vs-kp	3196	3196	36	2.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	31	credit-g	1000	1000	20	2.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	1501	semeion	1593	1593	256	10.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	11	balance-scale	625	625	4	3.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	1485	madelon	2600	2600	500	2.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	307	vowel	990	990	12	11.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	40975	car	1728	1728	6	4.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	4134	Bioresponse	3751	3751	1776	2.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	1461	bank-marketing	30000	45211	16	2.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	40966	MiceProtein	1080	1080	77	8.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	28	optdigits	5620	5620	64	10.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	1497	wall-robot-navigation	5456	5456	24	4.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	1478	har	10299	10299	561	6.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	40668	connect-4	30000	67557	42	3.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	41027	jungle_chess_2pcs_raw_endgame_complete	30000	44819	6	3.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	300	isolet	7797	7797	617	26.000	0.000	0.000	1	0.000	0.000	CLEAN

task	data_id	name	n_rows	n_raw	n_features	n_classes	dup_row_frac	conflict_row_frac	max_group	pred_exposure	obs_exposure	set
classification	40701	churn	5000	5000	20	2.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	6332	cylinder-bands	540	540	37	2.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	29	credit-approval	690	690	15	2.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	40982	steel-plates-fault	1941	1941	27	7.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	23517	numerai28.6	30000	96320	21	2.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	32	pendigits	10992	10992	16	10.000	0.000	0.000	1	0.000	0.000	CLEAN
regression	44984	cps88wages	28155	28155	6		0.898	0.898	71	0.853	0.853	DUP
regression	44966	solar_flare	1066	1066	10		0.894	0.665	73	0.848	0.854	DUP
regression	44964	superconductivity	21263	21263	81		0.400	0.396	110	0.334	0.334	DUP
regression	44971	white_wine	4898	4898	11		0.349	0.000	8	0.232	0.234	DUP
regression	44972	red_wine	1599	1599	11		0.288	0.000	4	0.181	0.181	DUP
regression	44970	QSAR_fish_toxicity	908	908	6		0.239	0.239	7	0.180	0.186	DUP
regression	44976	sarcos	30000	48933	21		0.182	0.000	2	0.109	0.109	DUP
regression	44993	health_insurance	22272	22272	11		0.124	0.088	19	0.096	0.097	DUP
regression	44990	brazilian_houses	10692	10692	9		0.107	0.058	24	0.081	0.083	DUP
regression	44963	physiochemical_protein	30000	45730	9		0.073	0.002	9	0.048	0.047	DUP
regression	44959	concrete_compressive_strength	1030	1030	8		0.051	0.019	4	0.041	0.040	DUP
regression	44962	forest_fires	517	517	12		0.048	0.033	3	0.030	0.030	DUP
regression	44979	diamonds	30000	53940	9		0.013	0.007	5	0.008	0.008	LOW
regression	44994	cars	804	804	17		0.002	0.002	2	0.001	0.002	LOW
regression	44974	video_transcoding	30000	68784	18		0.002	0.002	2	0.001	0.001	LOW
regression	44983	miami_housing	13932	13932	15		0.001	0.001	3	0.001	0.001	LOW
regression	44989	kings_county	21613	21613	21		0.000	0.000	2	0.000	0.000	LOW
regression	44987	socmob	1156	1156	5		0.000	0.000	1	0.000	0.000	CLEAN
regression	44992	fps_benchmark	24624	24624	43		0.000	0.000	1	0.000	0.000	CLEAN
regression	44973	grid_stability	10000	10000	12		0.000	0.000	1	0.000	0.000	CLEAN
regression	44965	geographical_origin_of_music	1059	1059	116		0.000	0.000	1	0.000	0.000	CLEAN
regression	44958	auction_verification	2043	2043	7		0.000	0.000	1	0.000	0.000	CLEAN
regression	45402	space_ga	3107	3107	6		0.000	0.000	1	0.000	0.000	CLEAN
regression	44957	airfoil_self_noise	1503	1503	5		0.000	0.000	1	0.000	0.000	CLEAN
regression	44969	naval_propulsion_plant	11934	11934	14		0.000	0.000	1	0.000	0.000	CLEAN
regression	44960	energy_efficiency	768	768	8		0.000	0.000	1	0.000	0.000	CLEAN
regression	44980	kin8nm	8192	8192	8		0.000	0.000	1	0.000	0.000	CLEAN
regression	44977	california_housing	20640	20640	8		0.000	0.000	1	0.000	0.000	CLEAN
regression	45012	fifa	19178	19178	28		0.000	0.000	1	0.000	0.000	CLEAN
regression	44956	abalone	4177	4177	8		0.000	0.000	1	0.000	0.000	CLEAN
regression	41021	Moneyball	1232	1232	14		0.000	0.000	1	0.000	0.000	CLEAN
regression	44981	pumadyn32nh	8192	8192	32		0.000	0.000	1	0.000	0.000	CLEAN
regression	44978	cpu_activity	8192	8192	21		0.000	0.000	1	0.000	0.000	CLEAN
regression	44975	wave_energy	30000	72000	48		0.000	0.000	1	0.000	0.000	CLEAN
regression	44967	student_performance_por	649	649	30		0.000	0.000	1	0.000	0.000	CLEAN

S2 Protocol deviations and post-hoc analyses

Freeze commit: `0e58bdd`. Everything below happened **after** the freeze and is disclosed in the manuscript.

#	When	What	Why	Effect on confirmatory results
1	Pipeline debugging, after the freeze but before full runs	`run_injection` / `run_capacity` now create their output directories (a bug fix found in quick mode)	Crash in quick mode	None
2	After the main results (primary H1 **not** supported at the median level)	Added **exploratory** analyses: (a) exposure strata π < 0.1, 0.1–0.3, ≥ 0.3; (b) the Proposition 1 decomposition of each dataset × learner gap into a seen-over-coverage term and a marginal term; (c) counts of dataset × learner pairs with gaps below −0.02 and −0.05; (d) Fig. 8	To explain *where* the gap arises. The theory says exposure alone is not sufficient: seen rows must also be over-covered. These analyses test that consequence	None. All pre-registered tests are reported unchanged and the new analyses are labelled exploratory
3	After the main results | **Verification run** on solar_flare (OpenML 44966): 200 extra random splits (seeds 1000–1199), ridge and LightGBM, absolute-residual score, α = 0.1 (`code/experiments/verify_solar_flare.py`)	With 10 seeds, MCP novel coverage on this dataset was 0.84–0.88, apparently contradicting Proposition 3	Confirms that it was sampling noise: MCP novel coverage 0.909 (95% CI 0.899–0.918) for LightGBM and 0.907 (0.896–0.917) for ridge; SCP novel coverage 0.78 and 0.77. Per-split SD ≈ 0.07 with about 36 novel calibration rows. Reported as a verification and a stated limitation (high split-to-split variability when novel calibration sets are small)	
4	After review round 1 (Reviews 1 and 5: suites are curated, and CTR23 excludes grouped data), **before any outcome on these data** | **External validation set**, specified in advance: every NASA MDP defect dataset on OpenML not in CC18 with ≥ 300 rows (mc1 #1056, pc2 #1069, kc3 #1065, mw1 #1071), plus KDDCup99 (10% version, #1113; documented heavy redundancy, Tavallaee et al. 2009). Same configuration as the main grid (10 seeds, 5 learners, both protocols). KDDCup99 (494,020 rows) is **uniformly row-subsampled** to 30,000 rows (seed 2026), because group-preserving subsampling cannot keep groups larger than the cap. Pre-specified test: novel-record gap of SCP-R per dataset and learner (LAC, α = 0.1), with seed-level 95% t-intervals; and the MCP-R novel gap	Tests whether the mechanism appears where duplication is known to be common	Additional evidence only; the confirmatory suite analysis is unchanged	
5	Review round 1 (Review 7)	Experiment G (computational cost of the audit and of MCP vs SCP), pre-registered but not yet run, is run now	Omission	None
6	Review round 2 (re-targeting to Pattern Recognition; anticipated baseline request), **before outcomes** | **Normalized-residual baseline** (Papadopoulos et al. 2011): regression DUP datasets (12), RF and LightGBM, 10 seeds, α = 0.1. Score $\lvert y - \hat{y} \rvert / (\hat{\sigma}(x) + \beta)$, where $\hat{\sigma}(x)$ is the mean distance to the k = 10 nearest training rows in the standardized feature space and β is the median of $\hat{\sigma}$ over calibration rows. Compared as SCP-R and MCP-R with the same score. Pre-specified question: does the adaptive score alone remove the novel-record gap?	A locally adaptive score shrinks intervals near training copies and might absorb the effect	Additional baseline; confirmatory results unchanged	

No dataset, seed or learner was excluded after outcomes were seen. No hyperparameter was changed.

S3a Main results on DUP datasets, α = 0.05

Medians over datasets of seed-averaged metrics; split CP with a random split (scp) and novelty-Mondrian CP (mcp).

task	model	score	n_datasets	scp_all_cov_median	scp_seen_cov_median	scp_novel_cov_median	scp_gap_median	scp_gap_min	p_H1_holm	spearman_exposure_gap	p_spearman	mcp_all_cov_median	mcp_novel_cov_median	mcp_gap_ci_lo	p_H2_holm	size_ratio_novel_median	size_ratio_seen_median	size_ratio_all_median
classification	logreg	lac	24	0.951	0.974	0.948	-0.002	-0.027	1.000	-0.551	0.005	0.952	0.952	-0.001	0.026	1.007	0.992	1.010
classification	logreg	aps	24	0.953	0.962	0.951	0.001	-0.011	1.000	-0.527	0.008	0.953	0.953	0.000	0.049	1.003	1.002	1.002
classification	rf	lac	24	0.952	0.990	0.947	-0.003	-0.026	1.000	-0.403	0.051	0.953	0.952	0.000	0.032	1.007	0.980	1.007

task	model	score	n_datasets	scp_all_cov_median	scp_seen_cov_median	scp_novel_cov_median	scp_gap_median	scp_gap_min	p_H1_holm	spearman_exposure_gap	p_spearman	mcp_all_cov_median	mcp_novel_cov_median	mcp_gap_ci_lo	p_H2_holm	size_ratio_novel_median	size_ratio_seen_median	size_ratio_all_median
classification	rf	aps	24	0.952	0.949	0.954	0.004	-0.011	1.000	-0.049	0.821	0.954	0.952	-0.000	0.996	0.993	1.107	1.006
classification	lgbm	lac	24	0.951	0.983	0.946	-0.004	-0.040	0.019	-0.598	0.002	0.952	0.951	-0.001	0.000	1.006	0.984	1.007
classification	lgbm	aps	24	0.952	0.966	0.949	-0.001	-0.032	1.000	-0.707	0.000	0.952	0.952	-0.001	0.004	1.006	1.000	1.005
classification	knn	lac	24	0.973	0.995	0.970	0.020	-0.026	1.000	-0.328	0.118	0.975	0.974	0.020	0.031	1.000	0.982	1.004
classification	knn	aps	24	0.957	0.980	0.954	0.004	-0.010	1.000	-0.118	0.582	0.959	0.957	0.002	0.022	1.002	0.999	1.001
classification	mlp	lac	24	0.950	0.977	0.947	-0.003	-0.029	1.000	-0.437	0.033	0.951	0.951	-0.001	0.022	1.009	0.997	1.007
classification	mlp	aps	24	0.952	0.965	0.950	0.000	-0.027	1.000	-0.409	0.047	0.953	0.952	0.000	0.026	1.005	0.999	1.002
regression	logreg	absres	12	0.949	0.960	0.948	-0.002	-0.106	1.000	-0.587	0.045	0.950	0.950	-0.004	0.077	1.017	0.948	1.005
regression	rf	absres	12	0.951	0.989	0.946	-0.004	-0.093	1.000	-0.636	0.026	0.952	0.952	-0.000	0.032	1.039	0.655	0.990
regression	lgbm	absres	12	0.950	0.986	0.946	-0.004	-0.102	1.000	-0.650	0.022	0.952	0.952	-0.001	0.041	1.043	0.789	0.999
regression	knn	absres	12	0.950	0.978	0.948	-0.002	-0.082	1.000	-0.636	0.026	0.952	0.952	-0.001	0.068	1.042	0.833	1.001
regression	mlp	absres	12	0.950	0.977	0.946	-0.004	-0.073	1.000	-0.622	0.031	0.950	0.950	-0.000	0.068	1.032	0.894	1.001

S3b Main results on DUP datasets, α = 0.1

Medians over datasets of seed-averaged metrics; split CP with a random split (scp) and novelty-Mondrian CP (mcp).

task	model	score	n_datasets	scp_all_cov_median	scp_seen_cov_median	scp_novel_cov_median	scp_gap_median	scp_gap_min	p_H1_holm	spearman_exposure_gap	p_spearman	mcp_all_cov_median	mcp_novel_cov_median	mcp_gap_ci_lo	p_H2_holm	size_ratio_novel_median	size_ratio_seen_median	size_ratio_all_median
classification	logreg	lac	24	0.901	0.937	0.895	-0.005	-0.065	0.343	-0.583	0.003	0.903	0.902	-0.000	0.001	1.009	0.962	1.008
classification	logreg	aps	24	0.900	0.913	0.899	-0.001	-0.015	1.000	-0.637	0.001	0.903	0.902	-0.001	0.107	1.005	0.997	1.004
classification	rf	lac	24	0.901	0.974	0.894	-0.006	-0.065	0.420	-0.414	0.044	0.903	0.901	-0.001	0.107	1.012	0.950	1.006
classification	rf	aps	24	0.903	0.891	0.905	0.005	-0.012	1.000	-0.118	0.582	0.903	0.901	-0.001	0.979	0.995	1.054	1.001
classification	lgbm	lac	24	0.900	0.968	0.894	-0.006	-0.067	0.092	-0.461	0.023	0.901	0.900	-0.002	0.005	1.011	0.928	1.004
classification	lgbm	aps	24	0.901	0.920	0.898	-0.002	-0.042	0.607	-0.838	0.000	0.901	0.902	-0.002	0.002	1.003	0.988	1.001
classification	knn	lac	24	0.934	0.978	0.926	0.026	-0.028	1.000	-0.356	0.088	0.938	0.936	0.026	0.061	1.000	0.975	1.009
classification	knn	aps	24	0.902	0.919	0.901	0.001	-0.016	1.000	-0.114	0.595	0.903	0.902	-0.001	0.229	1.001	0.992	1.001
classification	mlp	lac	24	0.900	0.947	0.896	-0.004	-0.069	0.395	-0.337	0.107	0.901	0.902	-0.001	0.001	1.012	0.939	1.005
classification	mlp	aps	24	0.901	0.909	0.900	0.000	-0.049	1.000	-0.628	0.001	0.903	0.903	0.000	0.002	1.005	0.990	1.005
regression	logreg	absres	12	0.901	0.920	0.898	-0.002	-0.168	1.000	-0.804	0.002	0.901	0.900	-0.005	0.154	1.026	0.938	1.001
regression	rf	absres	12	0.901	0.976	0.892	-0.008	-0.158	0.881	-0.685	0.014	0.900	0.900	-0.002	0.107	1.043	0.514	0.987
regression	lgbm	absres	12	0.900	0.965	0.894	-0.006	-0.169	1.000	-0.685	0.014	0.900	0.900	-0.001	0.061	1.027	0.782	0.994
regression	knn	absres	12	0.902	0.956	0.898	-0.002	-0.117	1.000	-0.371	0.236	0.902	0.900	-0.003	0.107	1.044	0.820	0.999
regression	mlp	absres	12	0.899	0.941	0.895	-0.005	-0.108	0.833	-0.622	0.031	0.900	0.899	-0.004	0.154	1.027	0.844	1.000
regression	cqr	cqr	12	0.903	0.965	0.894	-0.006	-0.037	0.881	-0.552	0.063	0.903	0.900	-0.001	0.011	1.017	0.954	1.006

S3c Main results on DUP datasets, α = 0.2

Medians over datasets of seed-averaged metrics; split CP with a random split (scp) and novelty-Mondrian CP (mcp).

task	model	score	n_datasets	scp_all_cov_median	scp_seen_cov_median	scp_novel_cov_median	scp_gap_median	scp_gap_min	p_H1_holm	spearman_exposure_gap	p_spearman	mcp_all_cov_median	mcp_novel_cov_median	mcp_gap_ci_lo	p_H2_holm	size_ratio_novel_median	size_ratio_seen_median	size_ratio_all_median
classification	logreg	lac	24	0.799	0.841	0.794	-0.006	-0.165	0.220	-0.326	0.120	0.803	0.803	-0.001	0.003	1.010	0.921	1.003
classification	logreg	aps	24	0.802	0.811	0.799	-0.001	-0.029	1.000	-0.516	0.010	0.803	0.801	-0.002	0.045	1.003	0.993	1.002
classification	rf	lac	24	0.801	0.941	0.787	-0.013	-0.164	0.297	-0.486	0.016	0.804	0.802	-0.003	0.036	1.020	0.882	1.004
classification	rf	aps	24	0.802	0.783	0.806	0.006	-0.021	1.000	-0.274	0.195	0.803	0.804	0.000	0.984	0.997	1.038	1.001
classification	lgbm	lac	24	0.801	0.932	0.789	-0.011	-0.150	0.044	-0.410	0.047	0.800	0.802	-0.003	0.006	1.019	0.869	1.002
classification	lgbm	aps	24	0.801	0.827	0.799	-0.001	-0.081	0.678	-0.677	0.000	0.802	0.803	-0.000	0.007	1.006	0.984	1.001
classification	knn	lac	24	0.849	0.942	0.839	0.039	-0.068	1.000	-0.449	0.028	0.864	0.858	0.044	0.039	1.000	0.966	1.000
classification	knn	aps	24	0.802	0.822	0.800	0.000	-0.034	1.000	-0.274	0.195	0.803	0.803	-0.002	0.045	1.007	0.993	1.002
classification	mlp	lac	24	0.801	0.862	0.792	-0.008	-0.154	0.363	-0.294	0.163	0.803	0.803	-0.001	0.003	1.013	0.895	1.007
classification	mlp	aps	24	0.801	0.818	0.798	-0.002	-0.051	1.000	-0.397	0.054	0.801	0.801	-0.001	0.045	1.003	0.991	1.002
regression	logreg	absres	12	0.802	0.825	0.797	-0.003	-0.258	1.000	-0.671	0.017	0.802	0.801	-0.005	0.151	1.005	0.955	1.001
regression	rf	absres	12	0.799	0.961	0.784	-0.016	-0.247	0.424	-0.566	0.055	0.799	0.799	-0.003	0.042	1.060	0.640	1.000
regression	lgbm	absres	12	0.800	0.912	0.790	-0.010	-0.253	1.000	-0.615	0.033	0.800	0.800	-0.003	0.045	1.037	0.774	1.000
regression	knn	absres	12	0.801	0.878	0.794	-0.006	-0.204	1.000	-0.594	0.042	0.802	0.801	-0.001	0.128	1.037	0.800	1.008
regression	mlp	absres	12	0.801	0.852	0.791	-0.009	-0.212	1.000	-0.636	0.026	0.801	0.800	-0.003	0.128	1.030	0.854	1.001

S4 Group-aware split (SCP-G), α = 0.1

task	model	score	n	g_cov_median	gap_median	ci_lo	ci_hi	noninferior_0_01	share_gap_lt_minus001
classification	logreg	lac	24	0.901	0.001	-0.002	0.004	True	0.125
classification	logreg	aps	24	0.899	-0.001	-0.003	0.002	True	0.042
classification	rf	lac	24	0.902	0.002	-0.001	0.006	True	0.000
classification	rf	aps	24	0.901	0.001	-0.002	0.004	True	0.042
classification	lgbm	lac	24	0.905	0.005	0.001	0.008	True	0.042
classification	lgbm	aps	24	0.901	0.001	-0.001	0.004	True	0.042
classification	knn	lac	24	0.933	0.033	0.030	0.041	True	0.000
classification	knn	aps	24	0.901	0.001	-0.001	0.007	True	0.000
classification	mlp	lac	24	0.902	0.002	-0.004	0.006	True	0.208
classification	mlp	aps	24	0.899	-0.001	-0.003	0.002	True	0.042
regression	logreg	absres	12	0.899	-0.001	-0.004	0.003	True	0.083
regression	rf	absres	12	0.898	-0.002	-0.004	0.001	True	0.000
regression	lgbm	absres	12	0.899	-0.001	-0.003	0.001	True	0.083
regression	knn	absres	12	0.902	0.002	-0.001	0.008	True	0.000
regression	mlp	absres	12	0.900	0.000	-0.003	0.002	True	0.083

task	model	score	n	g_cov_median	gap_median	ci_lo	ci_hi	noninferior_0_01	share_gap_lt_minus001
regression	cqr	cqr	12	0.900	0.000	-0.003	0.009	True	0.083

S5 Null control (CLEAN datasets), α = 0.1

task	model	score	n	scp_cov_median	scp_cov_mean	max_abs_mcp_minus_scp
classification	logreg	lac	32	0.900	0.901	0.000
classification	logreg	aps	32	0.901	0.901	0.000
classification	rf	lac	32	0.902	0.902	0.000
classification	rf	aps	32	0.901	0.900	0.000
classification	lgbm	lac	32	0.900	0.900	0.000
classification	lgbm	aps	32	0.901	0.902	0.000
classification	knn	lac	32	0.941	0.943	0.000
classification	knn	aps	32	0.900	0.905	0.000
classification	mlp	lac	32	0.902	0.902	0.000
classification	mlp	aps	32	0.901	0.901	0.000
regression	logreg	absres	18	0.899	0.900	0.000
regression	rf	absres	18	0.902	0.903	0.000
regression	lgbm	absres	18	0.902	0.902	0.000
regression	knn	absres	18	0.901	0.901	0.000
regression	mlp	absres	18	0.902	0.902	0.000
regression	cqr	cqr	18	0.901	0.901	0.000

S6 Friedman tests across learners (novel coverage, SCP-R, DUP)

family	n	chi2	p	median_logreg	median_rf	median_lgbm	median_knn	median_mlp
classification_lac	24	51.5667	0.0000	0.8954	0.8936	0.8936	0.9259	0.8956
classification_aps	24	16.5667	0.0023	0.8988	0.9052	0.8978	0.9009	0.9002
regression_absres	12	4.3333	0.3628	0.8978	0.8919	0.8938	0.8982	0.8946

S7 Exploratory exposure strata

stratum	score	n_pairs	n_datasets	gap_median	gap_p10	gap_min	share_lt_m002	share_lt_m005	mcp_gap_median	mcp_gap_min	mcp_share_lt_m002
pi<0.1	absres	25	5	-0.0000	-0.0047	-0.0093	0.0000	0.0000	0.0017	-0.0048	0.0000
pi<0.1	aps	60	12	0.0023	-0.0038	-0.0088	0.0000	0.0000	0.0033	-0.0127	0.0000
pi<0.1	cqr	5	5	0.0018	-0.0038	-0.0042	0.0000	0.0000	0.0062	-0.0037	0.0000
pi<0.1	lac	60	12	0.0012	-0.0101	-0.0153	0.0000	0.0000	0.0020	-0.0187	0.0000
0.1<=pi<0.3	absres	20	4	-0.0090	-0.0223	-0.0306	0.1500	0.0000	-0.0007	-0.0064	0.0000
0.1<=pi<0.3	aps	35	7	-0.0017	-0.0123	-0.0379	0.0286	0.0000	-0.0001	-0.0138	0.0000
0.1<=pi<0.3	cqr	4	4	-0.0153	-0.0216	-0.0225	0.2500	0.0000	-0.0011	-0.0019	0.0000
0.1<=pi<0.3	lac	35	7	-0.0053	-0.0276	-0.0444	0.2000	0.0000	0.0006	-0.0106	0.0000
pi>=0.3	absres	15	3	-0.0128	-0.1638	-0.1692	0.3333	0.3333	-0.0017	-0.0578	0.3333
pi>=0.3	aps	25	5	-0.0074	-0.0311	-0.0490	0.1600	0.0000	0.0064	-0.0194	0.0000
pi>=0.3	cqr	3	3	-0.0172	-0.0332	-0.0372	0.3333	0.0000	-0.0002	-0.0007	0.0000
pi>=0.3	lac	25	5	-0.0063	-0.0652	-0.0687	0.4400	0.3200	0.0117	-0.0070	0.0000

S8 Verification run on solar_flare (200 additional splits)

model	n_splits	scp_novel_mean	mcp_novel_mean	mcp_novel_pooled	mcp_mean_ci	mcp_per_split_sd
lgbm	200	0.7807	0.9086	0.9090	[0.8992, 0.9179]	0.0683
logreg	200	0.7746	0.9065	0.9076	[0.8963, 0.9165]	0.0754

S9 Duplicate injection: medians by rate

`gap`/`mcp_gap`: novel-record coverage − 0.9 for split / novelty-Mondrian CP; `limit_gap`: memorization limit −απ/(1−π).

family	rate	gap	mcp_gap	limit_gap	exposure	cov_seen
classification_knn	0.0000	0.0471	0.0471	-0.0000	0.0000	
classification_knn	0.1000	0.0297	0.0333	-0.0121	0.1081	0.9751
classification_knn	0.2500	0.0311	0.0458	-0.0317	0.2410	0.9713
classification_knn	0.5000	0.0175	0.0383	-0.0680	0.4046	0.9692
classification_knn	1.0000	0.0150	0.0366	-0.1484	0.5973	0.9658
classification_lgbm	0.0000	0.0000	0.0000	-0.0000	0.0000	
classification_lgbm	0.1000	-0.0128	-0.0004	-0.0121	0.1081	1.0000
classification_lgbm	0.2500	-0.0261	0.0009	-0.0317	0.2410	0.9996
classification_lgbm	0.5000	-0.0567	0.0002	-0.0680	0.4046	0.9953
classification_lgbm	1.0000	-0.0791	0.0023	-0.1484	0.5973	0.9610
classification_logreg	0.0000	-0.0006	-0.0006	-0.0000	0.0000	
classification_logreg	0.1000	-0.0025	0.0021	-0.0121	0.1081	0.9217
classification_logreg	0.2500	-0.0043	0.0011	-0.0317	0.2410	0.9112
classification_logreg	0.5000	-0.0054	0.0021	-0.0680	0.4046	0.9100
classification_logreg	1.0000	-0.0084	0.0007	-0.1484	0.5973	0.9049
classification_rf	0.0000	0.0013	0.0013	-0.0000	0.0000	
classification_rf	0.1000	-0.0117	-0.0002	-0.0121	0.1081	1.0000
classification_rf	0.2500	-0.0268	0.0016	-0.0317	0.2410	1.0000
classification_rf	0.5000	-0.0614	0.0030	-0.0680	0.4046	1.0000
classification_rf	1.0000	-0.1275	0.0035	-0.1484	0.5973	0.9994
regression_knn	0.0000	0.0026	0.0026	-0.0000	0.0000	

family	rate	gap	mcp_gap	limit_gap	exposure	cov_seen
regression_knn	0.1000	-0.0040	0.0002	-0.0121	0.1081	0.9314
regression_knn	0.2500	-0.0073	0.0016	-0.0316	0.2401	0.9383
regression_knn	0.5000	-0.0141	0.0030	-0.0673	0.4024	0.9326
regression_knn	1.0000	-0.0256	0.0034	-0.1484	0.5973	0.9208
regression_lgbm	0.0000	-0.0015	-0.0015	-0.0000	0.0000	
regression_lgbm	0.1000	-0.0078	0.0016	-0.0121	0.1081	0.9773
regression_lgbm	0.2500	-0.0237	-0.0009	-0.0316	0.2401	0.9759
regression_lgbm	0.5000	-0.0421	0.0016	-0.0673	0.4024	0.9546
regression_lgbm	1.0000	-0.0527	0.0045	-0.1484	0.5973	0.9378
regression_logreg	0.0000	0.0010	0.0010	-0.0000	0.0000	
regression_logreg	0.1000	0.0016	0.0017	-0.0121	0.1081	0.9084
regression_logreg	0.2500	0.0006	0.0024	-0.0316	0.2401	0.9049
regression_logreg	0.5000	0.0035	0.0044	-0.0673	0.4024	0.9002
regression_logreg	1.0000	-0.0007	0.0005	-0.1484	0.5973	0.9022
regression_rf	0.0000	0.0035	0.0035	-0.0000	0.0000	
regression_rf	0.1000	-0.0116	-0.0001	-0.0121	0.1081	0.9955
regression_rf	0.2500	-0.0286	0.0004	-0.0316	0.2401	0.9921
regression_rf	0.5000	-0.0535	0.0019	-0.0673	0.4024	0.9898
regression_rf	1.0000	-0.1179	0.0049	-0.1484	0.5973	0.9803

S10 Capacity: medians by level

`gap`/`mcp_gap`: novel-record coverage − 0.9 for split / novelty-Mondrian CP; `limit_gap`: memorization limit −απ/(1−π).

family	level	gap	mcp_gap	limit_gap	exposure	cov_seen
classification_knn	0	0.1000	0.1000	-0.0137	0.1207	1.0000
classification_knn	1	0.0241	0.0367	-0.0137	0.1207	0.9760
classification_knn	2	0.0001	0.0108	-0.0137	0.1207	0.9523
classification_knn	3	-0.0032	0.0046	-0.0137	0.1207	0.9388
classification_logreg	0	-0.0087	0.0010	-0.0137	0.1207	0.9493
classification_logreg	1	-0.0083	-0.0005	-0.0137	0.1207	0.9434
classification_logreg	2	-0.0030	-0.0004	-0.0137	0.1207	0.9340
classification_rf	0	-0.0072	0.0022	-0.0137	0.1207	0.9673
classification_rf	1	-0.0048	0.0002	-0.0137	0.1207	0.9555
classification_rf	2	-0.0083	0.0031	-0.0137	0.1207	0.9357
classification_rf	3	-0.0071	-0.0006	-0.0137	0.1207	0.9297
regression_knn	0	-0.0002	0.0050	-0.0243	0.1956	0.9792
regression_knn	1	-0.0033	0.0007	-0.0243	0.1956	0.9509
regression_knn	2	-0.0045	-0.0015	-0.0243	0.1956	0.9391
regression_knn	3	-0.0030	-0.0008	-0.0243	0.1956	0.9329
regression_logreg	0	-0.0051	0.0010	-0.0243	0.1956	0.9201
regression_logreg	1	-0.0065	0.0010	-0.0243	0.1956	0.9189
regression_logreg	2	-0.0073	-0.0001	-0.0243	0.1956	0.9133
regression_rf	0	-0.0132	-0.0015	-0.0243	0.1956	0.9723
regression_rf	1	-0.0114	-0.0016	-0.0243	0.1956	0.9628
regression_rf	2	-0.0028	-0.0007	-0.0243	0.1956	0.9311
regression_rf	3	-0.0061	-0.0008	-0.0243	0.1956	0.9208

S10b Normalized-residual baseline (regression DUP datasets, alpha = 0.1)

Mean over 10 splits. Score |y - yhat| / (sigma(x) + beta), sigma = mean distance to 10 nearest training rows.

data_id	name	model	method	cov_novel	cov_seen	cov_all	exposure	size_novel_norm	size_all_norm
44959	concrete_compressive_strength	lgbm	MCP	0.9188	0.9875	0.9214	0.0403	0.9513	inf
44959	concrete_compressive_strength	lgbm	SCP	0.9192	0.9607	0.9209	0.0403	0.9538	0.9437
44959	concrete_compressive_strength	rf	MCP	0.9020	0.9875	0.9053	0.0403	1.0618	inf
44959	concrete_compressive_strength	rf	SCP	0.9056	0.9607	0.9078	0.0403	1.0744	1.0631
44962	forest_fires	lgbm	MCP	0.9325	1.0000	0.9346	0.0298	2.3814	inf
44962	forest_fires	lgbm	SCP	0.9325	1.0000	0.9346	0.0298	2.3673	2.3615
44962	forest_fires	rf	MCP	0.9356	1.0000	0.9375	0.0298	1.9488	inf
44962	forest_fires	rf	SCP	0.9346	1.0000	0.9365	0.0298	1.9335	1.9278
44963	physiochemical_protein	lgbm	MCP	0.8983	0.9026	0.8985	0.0473	2.2672	2.2374
44963	physiochemical_protein	lgbm	SCP	0.8957	0.9527	0.8984	0.0473	2.2488	2.2337
44963	physiochemical_protein	rf	MCP	0.8988	0.9017	0.8990	0.0473	2.1199	2.0499
44963	physiochemical_protein	rf	SCP	0.8939	0.9894	0.8984	0.0473	2.0798	2.0658
44964	superconductivity	lgbm	MCP	0.9009	0.9003	0.9007	0.3335	1.6095	1.4871
44964	superconductivity	lgbm	SCP	0.9066	0.8876	0.9002	0.3335	1.6661	1.4936
44964	superconductivity	rf	MCP	0.9008	0.9024	0.9013	0.3335	1.4951	1.4265
44964	superconductivity	rf	SCP	0.9115	0.8788	0.9006	0.3335	1.6094	1.4426
44966	solar_flare	lgbm	MCP	0.8500	0.9058	0.8986	0.8535	3.8773	2.4263
44966	solar_flare	lgbm	SCP	0.8976	0.9015	0.9019	0.8535	4.5336	2.3932
44966	solar_flare	rf	MCP	0.8642	0.9023	0.8981	0.8535	4.3156	2.8043
44966	solar_flare	rf	SCP	0.9069	0.8909	0.8939	0.8535	5.2725	2.7854
44970	QSAR_fish_toxicity	lgbm	MCP	0.9074	0.9433	0.9133	0.1856	2.2286	2.0839
44970	QSAR_fish_toxicity	lgbm	SCP	0.8981	0.9610	0.9094	0.1856	2.1365	2.0188
44970	QSAR_fish_toxicity	rf	MCP	0.8968	0.9260	0.9011	0.1856	2.1146	1.9655

data_id	name	model	method	cov_novel	cov_seen	cov_all	exposure	size_novel_norm	size_all_norm
44970	QSAR_fish_toxicity	rf	SCP	0.8885	0.9580	0.9017	0.1856	2.0188	1.9084
44971	white_wine	lgbm	MCP	0.9031	0.8872	0.8991	0.2338	2.6822	2.3386
44971	white_wine	lgbm	SCP	0.8743	0.9895	0.9012	0.2338	2.4438	2.3782
44971	white_wine	rf	MCP	0.8967	0.8910	0.8952	0.2338	2.5892	2.1740
44971	white_wine	rf	SCP	0.8657	0.9992	0.8968	0.2338	2.3475	2.2845
44972	red_wine	lgbm	MCP	0.9080	0.8909	0.9050	0.1806	2.9832	2.5499
44972	red_wine	lgbm	SCP	0.8809	1.0000	0.9025	0.1806	2.7420	2.6989
44972	red_wine	rf	MCP	0.9026	0.9004	0.9022	0.1806	2.8203	2.4780
44972	red_wine	rf	SCP	0.8863	1.0000	0.9069	0.1806	2.6533	2.6117
44976	sarcos	lgbm	MCP	0.8995	0.9095	0.9006	0.1090	0.5241	0.5125
44976	sarcos	lgbm	SCP	0.8961	0.9395	0.9008	0.1090	0.5158	0.5122
44976	sarcos	rf	MCP	0.8993	0.8979	0.8992	0.1090	0.5470	0.5097
44976	sarcos	rf	SCP	0.8869	0.9931	0.8985	0.1090	0.5187	0.5151
44984	cps88wages	lgbm	MCP	0.8965	0.9017	0.9010	0.8527	3.0157	3.7609
44984	cps88wages	lgbm	SCP	0.9988	0.8835	0.9005	0.8527	19.9767	5.9962
44984	cps88wages	rf	MCP	0.8982	0.9019	0.9013	0.8527	3.4465	3.9701
44984	cps88wages	rf	SCP	0.9987	0.8843	0.9011	0.8527	20.9491	6.2874
44990	brazilian_houses	lgbm	MCP	0.9033	0.9022	0.9032	0.0829	0.6377	0.6131
44990	brazilian_houses	lgbm	SCP	0.9004	0.9353	0.9033	0.0829	0.6279	0.6117
44990	brazilian_houses	rf	MCP	0.8995	0.9001	0.8995	0.0829	0.6119	0.5870
44990	brazilian_houses	rf	SCP	0.8970	0.9265	0.8994	0.0829	0.6044	0.5885
44993	health_insurance	lgbm	MCP	0.9026	0.9021	0.9026	0.0970	3.0965	3.0631
44993	health_insurance	lgbm	SCP	0.9124	0.7915	0.9007	0.0970	3.2024	3.0830
44993	health_insurance	rf	MCP	0.9031	0.9024	0.9030	0.0970	3.2356	3.2470
44993	health_insurance	rf	SCP	0.9165	0.7597	0.9013	0.0970	3.4218	3.2943

S11 Computational cost (Experiment G)

task	data_id	name	n_rows	n_features	t_keys_s	t_audit_s	t_taxonomy_s	t_scp_threshold_s	t_mcp_threshold_s	t_lgbm_fit_predict_s
classification	1486	nomao	30000	118	0.1647	0.0073	0.0061	0.0000	0.0001	2.7280
classification	1590	adult	30000	14	0.0673	0.0056	0.0063	0.0000	0.0000	0.9187
classification	23517	numerai28.6	30000	21	0.0086	0.0077	0.0063	0.0000	0.0000	0.8931
classification	41027	jungle_chess_2pcs_raw_endgame_complete	30000	6	0.0029	0.0064	0.0061	0.0000	0.0000	1.7293
classification	40668	connect-4	30000	42	0.1556	0.0063	0.0067	0.0000	0.0000	2.7043
regression	44976	sarcos	30000	21	0.0117	0.0079	0.0060	0.0000	0.0001	1.3192
regression	44963	physiochemical_protein	30000	9	0.0040	0.0072	0.0063	0.0000	0.0001	0.6004
regression	44974	video_transcoding	30000	18	0.0199	0.0064	0.0058	0.0000	0.0000	0.7358
regression	44979	diamonds	30000	9	0.0179	0.0064	0.0056	0.0000	0.0000	0.6299
regression	44975	wave_energy	30000	48	0.0244	0.0074	0.0062	0.0000	0.0000	2.8234

Fig. S1

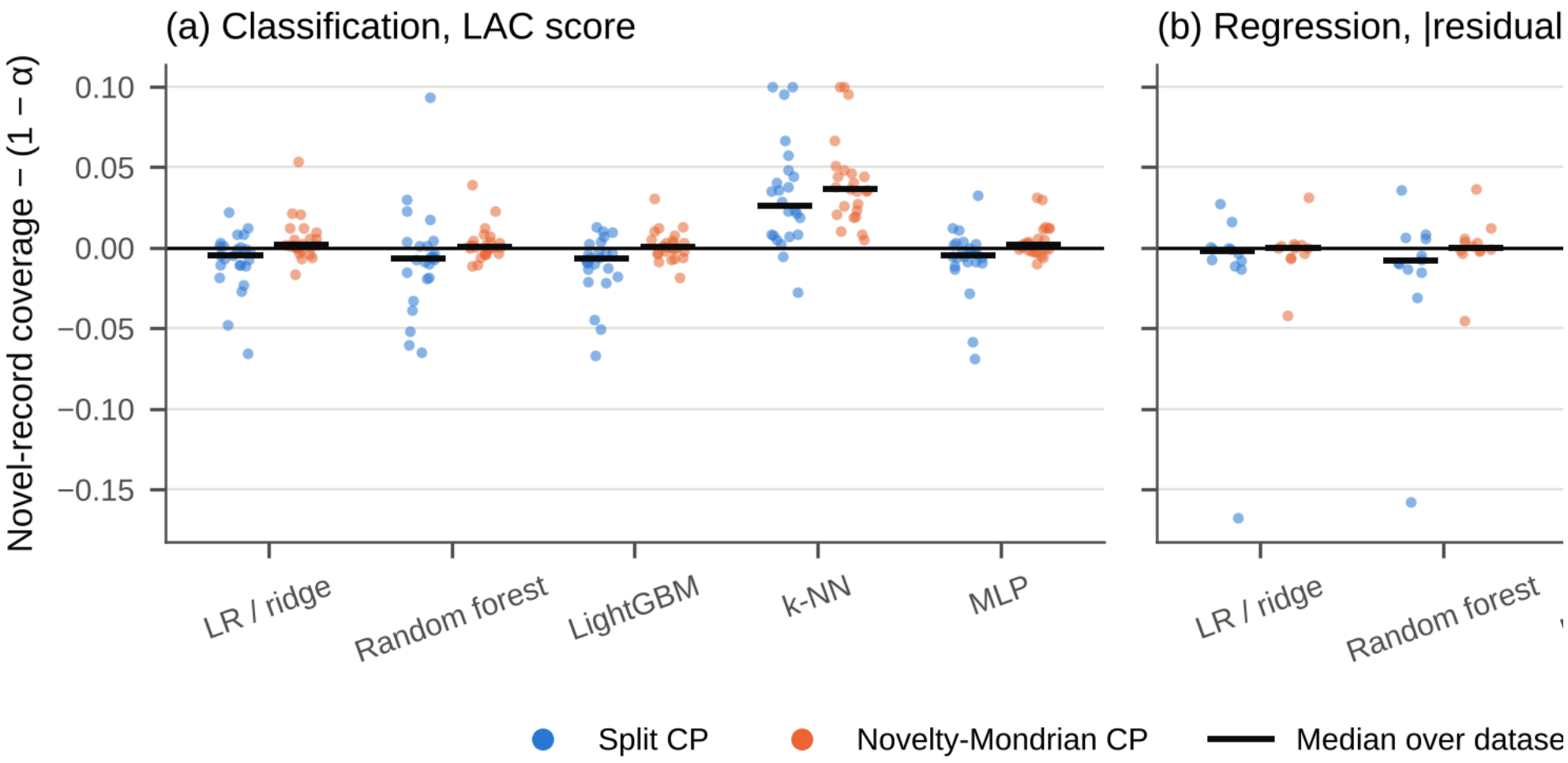


Fig. S1 Novel-record coverage gap per duplicated dataset (dots) and learner, split CP vs novelty-Mondrian CP, alpha = 0.1; bars: medians

S12 Computational environment

item	value
python	3.13.16 (main, Oct 1 2026, 15:40:24) [GCC 13.3.0]
platform	Linux-6.18.44-fc-v185-x86_64-with-glibc2.39
machine	x86_64
cpu_count	4
numpy	2.5.3
pandas	3.0.5
scipy	1.18.1
sklearn	1.9.1
lightgbm	4.7.0
time	2026-10-29 13:08:58

R1 Applications (all learners)

Means over 10 random splits, LAC score, α = 0.1. `pi`, `pi_p`: exposure and label-pure exposure; `predicted_gap`: Eq. (3); coverages and set sizes on new cases.

application	name	model	n_rows	pi	pi_p	predicted_gap	reported_cov	new_cov_standard	new_cov_fixed	seen_cov_standard	size_new_standard	size_new_fixed
Software defects	kc1	rf	2109	0.478	0.379	-0.061	0.903	0.835	0.907	0.977	1.235	1.482
Software defects	kc1	lgbm	2109	0.478	0.379	-0.061	0.904	0.833	0.903	0.980	1.251	1.504
Software defects	kc1	logreg	2109	0.478	0.379	-0.061	0.901	0.835	0.900	0.974	1.224	1.459
Software defects	kc2	rf	522	0.299	0.120	-0.014	0.893	0.867	0.896	0.951	1.263	1.333
Software defects	kc2	lgbm	522	0.299	0.120	-0.014	0.887	0.856	0.899	0.957	1.248	1.361
Software defects	kc2	logreg	522	0.299	0.120	-0.014	0.903	0.882	0.902	0.951	1.243	1.304
Software defects	pc1	rf	1109	0.167	0.160	-0.019	0.894	0.881	0.889	0.958	0.930	0.940
Software defects	pc1	lgbm	1109	0.167	0.160	-0.019	0.893	0.879	0.891	0.964	0.933	0.949
Software defects	pc1	logreg	1109	0.167	0.160	-0.019	0.901	0.895	0.901	0.930	0.956	0.964
Software defects	pc3	rf	1563	0.104	0.099	-0.011	0.899	0.893	0.900	0.956	1.006	1.020
Software defects	pc3	lgbm	1563	0.104	0.099	-0.011	0.901	0.894	0.903	0.962	1.016	1.037
Software defects	pc3	logreg	1563	0.104	0.099	-0.011	0.901	0.898	0.902	0.926	1.019	1.027
Software defects	pc4	rf	1458	0.096	0.094	-0.010	0.902	0.893	0.903	0.982	1.014	1.034
Software defects	pc4	lgbm	1458	0.096	0.094	-0.010	0.896	0.887	0.894	0.979	0.989	1.002
Software defects	pc4	logreg	1458	0.096	0.094	-0.010	0.915	0.908	0.912	0.975	1.012	1.024
Software defects	jm1	rf	10885	0.215	0.151	-0.018	0.898	0.882	0.894	0.955	1.278	1.330
Software defects	jm1	lgbm	10885	0.215	0.151	-0.018	0.894	0.878	0.897	0.952	1.277	1.344
Software defects	jm1	logreg	10885	0.215	0.151	-0.018	0.897	0.889	0.896	0.925	1.301	1.326
Software defects	mc1	rf	9466	0.819	0.811	-0.428	0.917	0.627	0.903	0.982	0.630	0.914
Software defects	mc1	lgbm	9466	0.819	0.811	-0.428	0.905	0.701	0.903	0.950	0.709	0.916
Software defects	mc1	logreg	9466	0.819	0.811	-0.428	0.901	0.725	0.898	0.940	0.735	0.916
Software defects	pc2	rf	5589	0.808	0.795	-0.388	0.910	0.625	0.894	0.978	0.632	0.909
Software defects	pc2	lgbm	5589	0.808	0.795	-0.388	0.898	0.728	0.896	0.939	0.739	0.910
Software defects	pc2	logreg	5589	0.808	0.795	-0.388	0.906	0.645	0.895	0.968	0.655	0.916
Software defects	kc3	rf	458	0.330	0.329	-0.049	0.931	0.898	0.945	0.996	1.100	1.245
Software defects	kc3	lgbm	458	0.330	0.329	-0.049	0.926	0.893	0.947	0.996	1.117	1.325
Software defects	kc3	logreg	458	0.330	0.329	-0.049	0.937	0.908	0.949	0.996	1.097	1.402
Software defects	mw1	rf	403	0.077	0.068	-0.007	0.895	0.896	0.900	0.916	0.999	1.004
Software defects	mw1	lgbm	403	0.077	0.068	-0.007	0.899	0.901	0.896	0.899	1.009	0.993
Software defects	mw1	logreg	403	0.077	0.068	-0.007	0.875	0.871	0.879	0.958	0.955	0.970
Network intrusion	KDDCup99	rf	494020	0.611	0.611	-0.157	0.955	0.886	0.916	0.998	0.886	0.916
Network intrusion	KDDCup99	lgbm	494020	0.611	0.611	-0.157	0.980	0.959	0.960	0.993	9.399	7.298
Network intrusion	KDDCup99	logreg	494020	0.611	0.611	-0.157	0.901	0.763	0.900	0.989	0.764	0.902
Phishing websites	PhishingWebsites	rf	11055	0.572	0.545	-0.120	0.899	0.848	0.901	0.937	0.855	0.916
Phishing websites	PhishingWebsites	lgbm	11055	0.572	0.545	-0.120	0.899	0.904	0.900	0.896	0.913	0.910
Phishing websites	PhishingWebsites	logreg	11055	0.572	0.545	-0.120	0.901	0.903	0.906	0.900	0.937	0.940

R2 Fresh validation set: selection log

Selection date 2026-10-29 14:36 UTC; pool of 163 OpenML data sets (studies 271, 269, 334–337). Inclusion used no labels and no models.

Included

data_id	name	task	n_raw	n_rows	subsampled	dup_row_frac	pred_exposure
181	yeast	classification	1484	1484	False	0.042	0.025
287	wine_quality	regression	6497	6497	False	0.334	0.220
416	yprop_4_1	regression	8885	8885	False	0.024	0.015
505	tecator	regression	240	240	False	0.167	0.100
1169	airlines	classification	539383	30000	True	0.922	0.709
4549	Buzzinsocialmedia_Twitter	regression	583250	30000	True	0.023	0.021
40498	wine-quality-white	classification	4898	4898	False	0.349	0.232
41540	black_friday	regression	166821	30000	True	0.724	0.633
42570	Mercedes_Benz_Greener_Manufacturing	regression	4209	4209	False	0.122	0.086
42732	sf-police-incidents	classification	2215023	30000	True	0.419	0.306
44055	analcatdata_supreme	regression	4052	4052	False	0.953	0.930
44065	nyc-taxi-green-dec-2016	regression	581835	30000	True	0.019	0.012
44068	particulate-matter-ukair-2017	regression	394299	30000	True	0.028	0.017
44125	MagicTelescope	classification	13376	13376	False	0.017	0.010
44142	Bike_Sharing_Demand	regression	17379	17379	False	0.045	0.027
44148	superconduct	regression	21263	21263	False	0.400	0.334
45022	Diabetes130US	classification	71090	30000	True	0.290	0.214
45026	heloc	classification	10000	10000	False	0.056	0.056
45039	compas-two-years	classification	4966	4966	False	0.738	0.654
45043	seattlecrime6	regression	52031	30000	True	0.999	0.999

Excluded (with reason)

data_id	name	reason
3		id already used in v1
12		id already used in v1
23		id already used in v1
31		id already used in v1
54		id already used in v1
188		id already used in v1

data_id	name	reason
201	pol	predicted exposure < 0.01
216	elevators	predicted exposure < 0.01
422	topo_2_1	predicted exposure < 0.01
507	space_ga	name already used in v1
531	boston	predicted exposure < 0.01
541	socmob	name already used in v1
546	sensory	predicted exposure < 0.01
550	quake	predicted exposure < 0.01
574	house_16H	predicted exposure < 0.01
1049		id already used in v1
1067		id already used in v1
1111	KDDCup09_appetency	predicted exposure < 0.01
1457	amazon-commerce-reviews	size limits
1461		id already used in v1
1464		id already used in v1
1468		id already used in v1
1475		id already used in v1
1486		id already used in v1
1487		id already used in v1
1489		id already used in v1
1494		id already used in v1
1515	micro-mass	predicted exposure < 0.01
1590		id already used in v1
1596	covertype	predicted exposure < 0.01
3050	QSAR-TID-11	download/parse failed: ValueError("Sparse ARFF datasets cannot be loaded with as_frame=Tru
3277	QSAR-TID-10980	download/parse failed: ValueError("Sparse ARFF datasets cannot be loaded with as_frame=Tru
4134		id already used in v1
4135	Amazon_employee_access	predicted exposure < 0.01
4534		id already used in v1
4538		id already used in v1
4541	Diabetes130US	predicted exposure < 0.01
23517		id already used in v1
40668		id already used in v1
40670		id already used in v1
40685	shuttle	predicted exposure < 0.01
40701		id already used in v1
40900	Satellite	predicted exposure < 0.01
40975		id already used in v1
40978		id already used in v1
40981	Australian	predicted exposure < 0.01
40982		id already used in v1
40983		id already used in v1
40984		id already used in v1
40996	Fashion-MNIST	predicted exposure < 0.01
41021		id already used in v1
41027		id already used in v1
41138	APSFailure	predicted exposure < 0.01
41142	christine	predicted exposure < 0.01
41143	jasmine	predicted exposure < 0.01
41144	madeline	predicted exposure < 0.01
41145	philippine	predicted exposure < 0.01
41146	sylvine	predicted exposure < 0.01
41147	albert	download/parse failed: OpenMLError('Dataset with data_id 41147 not found.')
41150	MiniBooNE	predicted exposure < 0.01
41156	ada	predicted exposure < 0.01
41157	arcene	size limits
41158	gina	predicted exposure < 0.01
41159	guillermo	size limits
41161	riccardo	size limits
41162	kick	predicted exposure < 0.01
41163	dilbert	size limits
41164	fabert	predicted exposure < 0.01
41165	robert	size limits
41166	volkert	predicted exposure < 0.01
41167	dionis	predicted exposure < 0.01
41168	jannis	predicted exposure < 0.01
41169	helena	predicted exposure < 0.01
41980	SAT11-HAND-runtime-regression	predicted exposure < 0.01
42225	diamonds	name already used in v1
42563	house_prices_nominal	predicted exposure < 0.01
42571	Allstate_Claims_Severity	predicted exposure < 0.01
42572	Santander_transaction_value	size limits
42688	Brazilian_houses	name already used in v1
42705	Yolanda	download/parse failed: ValueError('The md5 checksum of the file downloaded from https://op
42724	OnlineNewsPopularity	predicted exposure < 0.01
42726	abalone	name already used in v1

data_id	name	reason
42727	colleges	predicted exposure < 0.01
42728	Airlines_DepDelay_10M	size limits
42729	nyc-taxi-green-dec-2016	predicted exposure < 0.01
42730	us_crime	predicted exposure < 0.01
42731	house_sales	predicted exposure < 0.01
42733	Click_prediction_small	predicted exposure < 0.01
42734	okcupid-stem	predicted exposure < 0.01
42742	porto-seguro	predicted exposure < 0.01
42746	KDDCup99	name already used in v1
42769	Higgs	download/parse failed: ValueError('The md5 checksum of the file downloaded from https://op
43071	MIP-2016-regression	predicted exposure < 0.01
43072	KDDCup09-Upselling	size limits
44056	visualizing_soil	predicted exposure < 0.01
44059	diamonds	name already used in v1
44062	Brazilian_houses	name already used in v1
44063	Bike_Sharing_Demand	predicted exposure < 0.01
44066	house_sales	predicted exposure < 0.01
44069	SGEMM_GPU_kernel_performance	predicted exposure < 0.01
44089	credit	predicted exposure < 0.01
44120	electricity	name already used in v1
44121	covertype	predicted exposure < 0.01
44122	pol	predicted exposure < 0.01
44123	house_16H	predicted exposure < 0.01
44126	bank-marketing	name already used in v1
44128	MiniBooNE	predicted exposure < 0.01
44129	Higgs	predicted exposure < 0.01
44130	eye_movements	predicted exposure < 0.01
44132	cpu_act	predicted exposure < 0.01
44133	pol	predicted exposure < 0.01
44134	elevators	predicted exposure < 0.01
44137	Ailerons	predicted exposure < 0.01
44138	houses	predicted exposure < 0.01
44139	house_16H	predicted exposure < 0.01
44140	diamonds	name already used in v1
44141	Brazilian_houses	name already used in v1
44144	house_sales	predicted exposure < 0.01
44145	sulfur	predicted exposure < 0.01
44146	medical_charges	predicted exposure < 0.01
44147	MiamiHousing2016	predicted exposure < 0.01
44156	electricity	name already used in v1
44157	eye_movements	predicted exposure < 0.01
44159	covertype	predicted exposure < 0.01
45019	Bioresponse	name already used in v1
45020	default-of-credit-card-clients	predicted exposure < 0.01
45021	jannis	predicted exposure < 0.01
45028	california	predicted exposure < 0.01
45033	abalone	name already used in v1
45034	delays_zurich_transport	size limits
45035	albert	predicted exposure < 0.01
45036	default-of-credit-card-clients	predicted exposure < 0.01
45038	road-safety	predicted exposure < 0.01
45041	topo_2_1	predicted exposure < 0.01
45042	abalone	name already used in v1
45045	delays_zurich_transport	size limits
45046	Allstate_Claims_Severity	predicted exposure < 0.01
45047	Airlines_DepDelay_1M	predicted exposure < 0.01
45048	medical_charges	predicted exposure < 0.01
44061	Mercedes_Benz_Greener_Manufacturing	same data set already included under a lower id
44136	wine_quality	same data set already included under a lower id
44143	nyc-taxi-green-dec-2016	same data set already included under a lower id
45032	yprop_4_1	same data set already included under a lower id

R3 Fresh validation: coverage on new cases per data set

Means over 20 random splits, α = 0.1, LAC (classification) or absolute residual (regression). `bad`: share of splits with new-case coverage below 0.87; `fallback`: share of splits in which too few new calibration rows existed.

task	data_id	name	model	method	cov_novel	pi	pi_p	size_novel	bad	fallback
regression	44142	Bike_Sharing_Demand	lgbm	DCP	0.901	0.028	0.001	341.515	0.000	0.000
regression	44142	Bike_Sharing_Demand	lgbm	MCP	0.900	0.028	0.001	339.931	0.000	0.000
regression	44142	Bike_Sharing_Demand	lgbm	PAC_MCP	0.906	0.028	0.001	352.748	0.000	0.000
regression	44142	Bike_Sharing_Demand	lgbm	SCP	0.900	0.028	0.001	339.719	0.000	0.000
regression	44142	Bike_Sharing_Demand	logreg	DCP	0.902	0.028	0.001	454.310	0.000	0.000
regression	44142	Bike_Sharing_Demand	logreg	MCP	0.902	0.028	0.001	450.400	0.000	0.000
regression	44142	Bike_Sharing_Demand	logreg	PAC_MCP	0.908	0.028	0.001	463.901	0.000	0.000
regression	44142	Bike_Sharing_Demand	logreg	SCP	0.902	0.028	0.001	450.153	0.000	0.000
regression	44142	Bike_Sharing_Demand	rf	DCP	0.903	0.028	0.001	352.176	0.000	0.000

task	data_id	name	model	method	cov_novel	pi	pi_p	size_novel	bad	fallback
regression	44142	Bike_Sharing_Demand	rf	MCP	0.903	0.028	0.001	350.947	0.000	0.000
regression	44142	Bike_Sharing_Demand	rf	PAC_MCP	0.910	0.028	0.001	364.369	0.000	0.000
regression	44142	Bike_Sharing_Demand	rf	SCP	0.903	0.028	0.001	351.202	0.000	0.000
regression	4549	Buzzinsocialmedia_Twitter	lgbm	DCP	0.902	0.021	0.002	216.053	0.000	0.000
regression	4549	Buzzinsocialmedia_Twitter	lgbm	MCP	0.900	0.021	0.002	215.479	0.000	0.000
regression	4549	Buzzinsocialmedia_Twitter	lgbm	PAC_MCP	0.905	0.021	0.002	227.034	0.000	0.000
regression	4549	Buzzinsocialmedia_Twitter	lgbm	SCP	0.898	0.021	0.002	210.834	0.000	0.000
regression	4549	Buzzinsocialmedia_Twitter	logreg	DCP	0.902	0.021	0.002	266.688	0.000	0.000
regression	4549	Buzzinsocialmedia_Twitter	logreg	MCP	0.901	0.021	0.002	270.583	0.000	0.000
regression	4549	Buzzinsocialmedia_Twitter	logreg	PAC_MCP	0.906	0.021	0.002	286.324	0.000	0.000
regression	4549	Buzzinsocialmedia_Twitter	logreg	SCP	0.899	0.021	0.002	264.922	0.000	0.000
regression	4549	Buzzinsocialmedia_Twitter	rf	DCP	0.901	0.021	0.002	215.307	0.000	0.000
regression	4549	Buzzinsocialmedia_Twitter	rf	MCP	0.901	0.021	0.002	222.432	0.000	0.000
regression	4549	Buzzinsocialmedia_Twitter	rf	PAC_MCP	0.906	0.021	0.002	233.674	0.000	0.000
regression	4549	Buzzinsocialmedia_Twitter	rf	SCP	0.899	0.021	0.002	217.523	0.000	0.000
classification	45022	Diabetes130US	lgbm	DCP	0.900	0.214	0.087	1.691	0.000	0.000
classification	45022	Diabetes130US	lgbm	MCP	0.900	0.214	0.087	1.689	0.000	0.000
classification	45022	Diabetes130US	lgbm	PAC_MCP	0.905	0.214	0.087	1.703	0.000	0.000
classification	45022	Diabetes130US	lgbm	SCP	0.890	0.214	0.087	1.663	0.000	0.000
classification	45022	Diabetes130US	logreg	DCP	0.901	0.214	0.087	1.683	0.000	0.000
classification	45022	Diabetes130US	logreg	MCP	0.900	0.214	0.087	1.678	0.000	0.000
classification	45022	Diabetes130US	logreg	PAC_MCP	0.905	0.214	0.087	1.693	0.000	0.000
classification	45022	Diabetes130US	logreg	SCP	0.892	0.214	0.087	1.655	0.000	0.000
classification	45022	Diabetes130US	rf	DCP	0.900	0.214	0.087	1.725	0.000	0.000
classification	45022	Diabetes130US	rf	MCP	0.900	0.214	0.087	1.732	0.000	0.000
classification	45022	Diabetes130US	rf	PAC_MCP	0.906	0.214	0.087	1.745	0.000	0.000
classification	45022	Diabetes130US	rf	SCP	0.942	0.214	0.087	1.835	0.000	0.000
classification	44125	MagicTelescope	lgbm	DCP	0.898	0.010	0.010	1.104	0.000	0.000
classification	44125	MagicTelescope	lgbm	MCP	0.897	0.010	0.010	1.093	0.000	0.000
classification	44125	MagicTelescope	lgbm	PAC_MCP	0.905	0.010	0.010	1.113	0.000	0.000
classification	44125	MagicTelescope	lgbm	SCP	0.897	0.010	0.010	1.091	0.000	0.000
classification	44125	MagicTelescope	logreg	DCP	0.899	0.010	0.010	1.355	0.000	0.000
classification	44125	MagicTelescope	logreg	MCP	0.899	0.010	0.010	1.341	0.000	0.000
classification	44125	MagicTelescope	logreg	PAC_MCP	0.906	0.010	0.010	1.371	0.000	0.000
classification	44125	MagicTelescope	logreg	SCP	0.899	0.010	0.010	1.340	0.000	0.000
classification	44125	MagicTelescope	rf	DCP	0.901	0.010	0.010	1.119	0.000	0.000
classification	44125	MagicTelescope	rf	MCP	0.899	0.010	0.010	1.105	0.000	0.000
classification	44125	MagicTelescope	rf	PAC_MCP	0.906	0.010	0.010	1.125	0.000	0.000
classification	44125	MagicTelescope	rf	SCP	0.898	0.010	0.010	1.103	0.000	0.000
regression	42570	Mercedes_Benz_Greener_Manufacturing	lgbm	DCP	0.909	0.088	0.000	24.034	0.000	0.000
regression	42570	Mercedes_Benz_Greener_Manufacturing	lgbm	MCP	0.904	0.088	0.000	23.343	0.000	0.000
regression	42570	Mercedes_Benz_Greener_Manufacturing	lgbm	PAC_MCP	0.917	0.088	0.000	24.879	0.000	0.000
regression	42570	Mercedes_Benz_Greener_Manufacturing	lgbm	SCP	0.906	0.088	0.000	23.584	0.000	0.000
regression	42570	Mercedes_Benz_Greener_Manufacturing	logreg	DCP	0.909	0.088	0.000	23.888	0.000	0.000
regression	42570	Mercedes_Benz_Greener_Manufacturing	logreg	MCP	0.906	0.088	0.000	23.303	0.000	0.000
regression	42570	Mercedes_Benz_Greener_Manufacturing	logreg	PAC_MCP	0.916	0.088	0.000	24.993	0.000	0.000
regression	42570	Mercedes_Benz_Greener_Manufacturing	logreg	SCP	0.906	0.088	0.000	23.269	0.000	0.000
regression	42570	Mercedes_Benz_Greener_Manufacturing	rf	DCP	0.905	0.088	0.000	24.578	0.000	0.000
regression	42570	Mercedes_Benz_Greener_Manufacturing	rf	MCP	0.902	0.088	0.000	23.661	0.000	0.000
regression	42570	Mercedes_Benz_Greener_Manufacturing	rf	PAC_MCP	0.914	0.088	0.000	25.621	0.000	0.000
regression	42570	Mercedes_Benz_Greener_Manufacturing	rf	SCP	0.906	0.088	0.000	24.320	0.000	0.000
classification	1169	airlines	lgbm	DCP	0.897	0.711	0.354	1.600	0.000	0.000
classification	1169	airlines	lgbm	MCP	0.896	0.711	0.354	1.595	0.000	0.000
classification	1169	airlines	lgbm	PAC_MCP	0.904	0.711	0.354	1.618	0.000	0.000
classification	1169	airlines	lgbm	SCP	0.902	0.711	0.354	1.613	0.000	0.000
classification	1169	airlines	logreg	DCP	0.902	0.711	0.354	1.610	0.000	0.000
classification	1169	airlines	logreg	MCP	0.899	0.711	0.354	1.599	0.000	0.000
classification	1169	airlines	logreg	PAC_MCP	0.909	0.711	0.354	1.630	0.000	0.000
classification	1169	airlines	logreg	SCP	0.904	0.711	0.354	1.616	0.000	0.000
classification	1169	airlines	rf	DCP	0.900	0.711	0.354	1.630	0.000	0.000
classification	1169	airlines	rf	MCP	0.899	0.711	0.354	1.631	0.000	0.000
classification	1169	airlines	rf	PAC_MCP	0.908	0.711	0.354	1.657	0.000	0.000
classification	1169	airlines	rf	SCP	0.987	0.711	0.354	1.931	0.000	0.000
regression	44055	analcatdata_supreme	lgbm	DCP	0.927	0.934	0.809	inf	0.200	0.000
regression	44055	analcatdata_supreme	lgbm	MCP	0.891	0.934	0.809	0.644	0.350	0.000
regression	44055	analcatdata_supreme	lgbm	PAC_MCP	0.948	0.934	0.809	0.985	0.050	0.000
regression	44055	analcatdata_supreme	lgbm	SCP	0.567	0.934	0.809	0.054	1.000	0.000

task	data_id	name	model	method	cov_novel	pi	pi_p	size_novel	bad	fallback
regression	44055	analcatdata_supreme	logreg	DCP	0.939	0.934	0.809	inf	0.050	0.000
regression	44055	analcatdata_supreme	logreg	MCP	0.906	0.934	0.809	1.317	0.300	0.000
regression	44055	analcatdata_supreme	logreg	PAC_MCP	0.946	0.934	0.809	1.877	0.100	0.000
regression	44055	analcatdata_supreme	logreg	SCP	0.877	0.934	0.809	1.006	0.500	0.000
regression	44055	analcatdata_supreme	rf	DCP	0.928	0.934	0.809	inf	0.100	0.000
regression	44055	analcatdata_supreme	rf	MCP	0.893	0.934	0.809	0.747	0.250	0.000
regression	44055	analcatdata_supreme	rf	PAC_MCP	0.943	0.934	0.809	1.152	0.050	0.000
regression	44055	analcatdata_supreme	rf	SCP	0.625	0.934	0.809	0.042	1.000	0.000
regression	41540	black_friday	lgbm	DCP	0.900	0.631	0.000	inf	0.000	0.000
regression	41540	black_friday	lgbm	MCP	0.899	0.631	0.000	10967.095	0.000	0.000
regression	41540	black_friday	lgbm	PAC_MCP	0.908	0.631	0.000	11280.752	0.000	0.000
regression	41540	black_friday	lgbm	SCP	0.914	0.631	0.000	11463.504	0.000	0.000
regression	41540	black_friday	logreg	DCP	0.902	0.631	0.000	inf	0.000	0.000
regression	41540	black_friday	logreg	MCP	0.901	0.631	0.000	16424.208	0.000	0.000
regression	41540	black_friday	logreg	PAC_MCP	0.910	0.631	0.000	16764.098	0.000	0.000
regression	41540	black_friday	logreg	SCP	0.863	0.631	0.000	14972.797	0.650	0.000
regression	41540	black_friday	rf	DCP	0.900	0.631	0.000	inf	0.000	0.000
regression	41540	black_friday	rf	MCP	0.899	0.631	0.000	11704.629	0.000	0.000
regression	41540	black_friday	rf	PAC_MCP	0.908	0.631	0.000	12052.015	0.000	0.000
regression	41540	black_friday	rf	SCP	0.918	0.631	0.000	12478.025	0.000	0.000
classification	45039	compas-two-years	lgbm	DCP	0.897	0.654	0.206	1.588	0.150	0.000
classification	45039	compas-two-years	lgbm	MCP	0.896	0.654	0.206	1.571	0.200	0.000
classification	45039	compas-two-years	lgbm	PAC_MCP	0.914	0.654	0.206	1.625	0.050	0.000
classification	45039	compas-two-years	lgbm	SCP	0.868	0.654	0.206	1.488	0.450	0.000
classification	45039	compas-two-years	logreg	DCP	0.903	0.654	0.206	1.546	0.100	0.000
classification	45039	compas-two-years	logreg	MCP	0.896	0.654	0.206	1.509	0.200	0.000
classification	45039	compas-two-years	logreg	PAC_MCP	0.916	0.654	0.206	1.585	0.000	0.000
classification	45039	compas-two-years	logreg	SCP	0.857	0.654	0.206	1.390	0.700	0.000
classification	45039	compas-two-years	rf	DCP	0.900	0.654	0.206	1.612	0.100	0.000
classification	45039	compas-two-years	rf	MCP	0.894	0.654	0.206	1.592	0.250	0.000
classification	45039	compas-two-years	rf	PAC_MCP	0.913	0.654	0.206	1.650	0.000	0.000
classification	45039	compas-two-years	rf	SCP	0.924	0.654	0.206	1.678	0.000	0.000
classification	45026	heloc	lgbm	DCP	0.905	0.055	0.000	1.465	0.000	0.000
classification	45026	heloc	lgbm	MCP	0.904	0.055	0.000	1.461	0.000	0.000
classification	45026	heloc	lgbm	PAC_MCP	0.912	0.055	0.000	1.492	0.000	0.000
classification	45026	heloc	lgbm	SCP	0.897	0.055	0.000	1.438	0.000	0.000
classification	45026	heloc	logreg	DCP	0.905	0.055	0.000	1.477	0.000	0.000
classification	45026	heloc	logreg	MCP	0.905	0.055	0.000	1.472	0.000	0.000
classification	45026	heloc	logreg	PAC_MCP	0.913	0.055	0.000	1.505	0.000	0.000
classification	45026	heloc	logreg	SCP	0.899	0.055	0.000	1.450	0.000	0.000
classification	45026	heloc	rf	DCP	0.906	0.055	0.000	1.461	0.000	0.000
classification	45026	heloc	rf	MCP	0.905	0.055	0.000	1.458	0.000	0.000
classification	45026	heloc	rf	PAC_MCP	0.914	0.055	0.000	1.490	0.000	0.000
classification	45026	heloc	rf	SCP	0.899	0.055	0.000	1.436	0.000	0.000
regression	44065	nyc-taxi-green-dec-2016	lgbm	DCP	0.901	0.012	0.011	1.272	0.000	0.000
regression	44065	nyc-taxi-green-dec-2016	lgbm	MCP	0.901	0.012	0.011	1.202	0.000	0.000
regression	44065	nyc-taxi-green-dec-2016	lgbm	PAC_MCP	0.906	0.012	0.011	1.233	0.000	0.000
regression	44065	nyc-taxi-green-dec-2016	lgbm	SCP	0.899	0.012	0.011	1.195	0.000	0.000
regression	44065	nyc-taxi-green-dec-2016	logreg	DCP	0.902	0.012	0.011	1.743	0.000	0.000
regression	44065	nyc-taxi-green-dec-2016	logreg	MCP	0.900	0.012	0.011	1.670	0.000	0.000
regression	44065	nyc-taxi-green-dec-2016	logreg	PAC_MCP	0.905	0.012	0.011	1.691	0.000	0.000
regression	44065	nyc-taxi-green-dec-2016	logreg	SCP	0.900	0.012	0.011	1.666	0.000	0.000
regression	44065	nyc-taxi-green-dec-2016	rf	DCP	0.902	0.012	0.011	1.277	0.000	0.000
regression	44065	nyc-taxi-green-dec-2016	rf	MCP	0.901	0.012	0.011	1.245	0.000	0.000
regression	44065	nyc-taxi-green-dec-2016	rf	PAC_MCP	0.906	0.012	0.011	1.276	0.000	0.000
regression	44065	nyc-taxi-green-dec-2016	rf	SCP	0.900	0.012	0.011	1.238	0.000	0.000
regression	44068	particulate-matter-ukair-2017	lgbm	DCP	0.898	0.017	0.000	1.118	0.000	0.000
regression	44068	particulate-matter-ukair-2017	lgbm	MCP	0.897	0.017	0.000	1.117	0.000	0.000
regression	44068	particulate-matter-ukair-2017	lgbm	PAC_MCP	0.902	0.017	0.000	1.142	0.000	0.000
regression	44068	particulate-matter-ukair-2017	lgbm	SCP	0.898	0.017	0.000	1.119	0.000	0.000
regression	44068	particulate-matter-ukair-2017	logreg	DCP	0.899	0.017	0.000	1.326	0.000	0.000
regression	44068	particulate-matter-ukair-2017	logreg	MCP	0.900	0.017	0.000	1.321	0.000	0.000
regression	44068	particulate-matter-ukair-2017	logreg	PAC_MCP	0.904	0.017	0.000	1.348	0.000	0.000
regression	44068	particulate-matter-ukair-2017	logreg	SCP	0.900	0.017	0.000	1.321	0.000	0.000
regression	44068	particulate-matter-ukair-2017	rf	DCP	0.900	0.017	0.000	1.192	0.000	0.000
regression	44068	particulate-matter-ukair-2017	rf	MCP	0.899	0.017	0.000	1.191	0.000	0.000
regression	44068	particulate-matter-ukair-2017	rf	PAC_MCP	0.904	0.017	0.000	1.217	0.000	0.000
regression	44068	particulate-matter-ukair-2017	rf	SCP	0.900	0.017	0.000	1.197	0.000	0.000
regression	45043	seattlecrime6	lgbm	MCP	0.988	0.999	0.000	inf	0.050	0.750
regression	45043	seattlecrime6	lgbm	PAC_MCP	1.000	0.999	0.000	inf	0.000	1.000
regression	45043	seattlecrime6	lgbm	SCP	0.973	0.999	0.000	1297.812	0.100	0.000
regression	45043	seattlecrime6	logreg	MCP	0.979	0.999	0.000	inf	0.100	0.750
regression	45043	seattlecrime6	logreg	PAC_MCP	1.000	0.999	0.000	inf	0.000	1.000
regression	45043	seattlecrime6	logreg	SCP	0.995	0.999	0.000	1283.382	0.000	0.000
regression	45043	seattlecrime6	rf	MCP	1.000	0.999	0.000	inf	0.000	0.750
regression	45043	seattlecrime6	rf	PAC_MCP	1.000	0.999	0.000	inf	0.000	1.000

task	data_id	name	model	method	cov_novel	pi	pi_p	size_novel	bad	fallback
regression	45043	seattlecrime6	rf	SCP	0.967	0.999	0.000	1299.429	0.100	0.000
classification	42732	sf-police-incidents	lgbm	DCP	0.901	0.303	0.241	1.098	0.000	0.000
classification	42732	sf-police-incidents	lgbm	MCP	0.900	0.303	0.241	1.089	0.000	0.000
classification	42732	sf-police-incidents	lgbm	PAC_MCP	0.906	0.303	0.241	1.128	0.000	0.000
classification	42732	sf-police-incidents	lgbm	SCP	0.901	0.303	0.241	1.098	0.000	0.000
classification	42732	sf-police-incidents	logreg	DCP	0.901	0.303	0.241	1.089	0.000	0.000
classification	42732	sf-police-incidents	logreg	MCP	0.900	0.303	0.241	1.082	0.000	0.000
classification	42732	sf-police-incidents	logreg	PAC_MCP	0.906	0.303	0.241	1.121	0.000	0.000
classification	42732	sf-police-incidents	logreg	SCP	0.908	0.303	0.241	1.131	0.000	0.000
classification	42732	sf-police-incidents	rf	DCP	0.900	0.303	0.241	1.110	0.000	0.000
classification	42732	sf-police-incidents	rf	MCP	0.900	0.303	0.241	1.102	0.000	0.000
classification	42732	sf-police-incidents	rf	PAC_MCP	0.906	0.303	0.241	1.150	0.000	0.000
classification	42732	sf-police-incidents	rf	SCP	0.901	0.303	0.241	1.109	0.000	0.000
regression	44148	superconduct	lgbm	DCP	0.900	0.332	0.002	32.524	0.000	0.000
regression	44148	superconduct	lgbm	MCP	0.900	0.332	0.002	32.715	0.000	0.000
regression	44148	superconduct	lgbm	PAC_MCP	0.907	0.332	0.002	34.116	0.000	0.000
regression	44148	superconduct	lgbm	SCP	0.888	0.332	0.002	30.529	0.000	0.000
regression	44148	superconduct	logreg	DCP	0.899	0.332	0.002	58.628	0.000	0.000
regression	44148	superconduct	logreg	MCP	0.899	0.332	0.002	59.524	0.000	0.000
regression	44148	superconduct	logreg	PAC_MCP	0.905	0.332	0.002	60.749	0.000	0.000
regression	44148	superconduct	logreg	SCP	0.890	0.332	0.002	58.161	0.000	0.000
regression	44148	superconduct	rf	DCP	0.899	0.332	0.002	31.142	0.000	0.000
regression	44148	superconduct	rf	MCP	0.899	0.332	0.002	31.013	0.000	0.000
regression	44148	superconduct	rf	PAC_MCP	0.906	0.332	0.002	32.452	0.000	0.000
regression	44148	superconduct	rf	SCP	0.891	0.332	0.002	29.558	0.000	0.000
regression	505	tecator	lgbm	DCP	0.952	0.110	0.110	inf	0.050	0.000
regression	505	tecator	lgbm	MCP	0.934	0.110	0.110	10.444	0.100	0.000
regression	505	tecator	lgbm	PAC_MCP	0.963	0.110	0.110	14.716	0.050	0.000
regression	505	tecator	lgbm	SCP	0.934	0.110	0.110	10.444	0.100	0.000
regression	505	tecator	logreg	DCP	0.943	0.110	0.110	inf	0.100	0.000
regression	505	tecator	logreg	MCP	0.904	0.110	0.110	3.430	0.250	0.000
regression	505	tecator	logreg	PAC_MCP	0.964	0.110	0.110	5.646	0.000	0.000
regression	505	tecator	logreg	SCP	0.918	0.110	0.110	3.565	0.150	0.000
regression	505	tecator	rf	DCP	0.963	0.110	0.110	inf	0.050	0.000
regression	505	tecator	rf	MCP	0.913	0.110	0.110	6.871	0.150	0.000
regression	505	tecator	rf	PAC_MCP	0.961	0.110	0.110	8.756	0.000	0.000
regression	505	tecator	rf	SCP	0.913	0.110	0.110	6.871	0.150	0.000
classification	40498	wine-quality-white	lgbm	DCP	0.903	0.235	0.235	2.404	0.050	0.000
classification	40498	wine-quality-white	lgbm	MCP	0.901	0.235	0.235	2.369	0.050	0.000
classification	40498	wine-quality-white	lgbm	PAC_MCP	0.915	0.235	0.235	2.476	0.000	0.000
classification	40498	wine-quality-white	lgbm	SCP	0.873	0.235	0.235	2.162	0.400	0.000
classification	40498	wine-quality-white	logreg	DCP	0.900	0.235	0.235	2.403	0.000	0.000
classification	40498	wine-quality-white	logreg	MCP	0.898	0.235	0.235	2.365	0.000	0.000
classification	40498	wine-quality-white	logreg	PAC_MCP	0.914	0.235	0.235	2.492	0.000	0.000
classification	40498	wine-quality-white	logreg	SCP	0.900	0.235	0.235	2.384	0.000	0.000
classification	40498	wine-quality-white	rf	DCP	0.905	0.235	0.235	2.350	0.000	0.000
classification	40498	wine-quality-white	rf	MCP	0.904	0.235	0.235	2.313	0.050	0.000
classification	40498	wine-quality-white	rf	PAC_MCP	0.919	0.235	0.235	2.433	0.000	0.000
classification	40498	wine-quality-white	rf	SCP	0.875	0.235	0.235	2.110	0.400	0.000
regression	287	wine_quality	lgbm	DCP	0.902	0.217	0.217	2.294	0.000	0.000
regression	287	wine_quality	lgbm	MCP	0.900	0.217	0.217	2.283	0.000	0.000
regression	287	wine_quality	lgbm	PAC_MCP	0.913	0.217	0.217	2.384	0.000	0.000
regression	287	wine_quality	lgbm	SCP	0.872	0.217	0.217	2.093	0.350	0.000
regression	287	wine_quality	logreg	DCP	0.901	0.217	0.217	2.381	0.000	0.000
regression	287	wine_quality	logreg	MCP	0.899	0.217	0.217	2.348	0.000	0.000
regression	287	wine_quality	logreg	PAC_MCP	0.910	0.217	0.217	2.461	0.000	0.000
regression	287	wine_quality	logreg	SCP	0.901	0.217	0.217	2.368	0.000	0.000
regression	287	wine_quality	rf	DCP	0.905	0.217	0.217	2.273	0.000	0.000
regression	287	wine_quality	rf	MCP	0.902	0.217	0.217	2.260	0.000	0.000
regression	287	wine_quality	rf	PAC_MCP	0.913	0.217	0.217	2.357	0.000	0.000
regression	287	wine_quality	rf	SCP	0.876	0.217	0.217	2.049	0.300	0.000
classification	181	yeast	lgbm	DCP	0.901	0.025	0.025	3.006	0.100	0.000
classification	181	yeast	lgbm	MCP	0.892	0.025	0.025	2.745	0.150	0.000
classification	181	yeast	lgbm	PAC_MCP	0.916	0.025	0.025	3.126	0.000	0.000
classification	181	yeast	lgbm	SCP	0.890	0.025	0.025	2.724	0.200	0.000
classification	181	yeast	logreg	DCP	0.899	0.025	0.025	2.526	0.150	0.000
classification	181	yeast	logreg	MCP	0.895	0.025	0.025	2.396	0.100	0.000
classification	181	yeast	logreg	PAC_MCP	0.917	0.025	0.025	2.705	0.000	0.000
classification	181	yeast	logreg	SCP	0.894	0.025	0.025	2.384	0.100	0.000
classification	181	yeast	rf	DCP	0.903	0.025	0.025	2.671	0.100	0.000
classification	181	yeast	rf	MCP	0.900	0.025	0.025	2.490	0.150	0.000
classification	181	yeast	rf	PAC_MCP	0.917	0.025	0.025	2.754	0.050	0.000
classification	181	yeast	rf	SCP	0.898	0.025	0.025	2.465	0.150	0.000
regression	416	yprop_4_1	lgbm	DCP	0.900	0.015	0.000	0.095	0.000	0.000
regression	416	yprop_4_1	lgbm	MCP	0.898	0.015	0.000	0.094	0.000	0.000
regression	416	yprop_4_1	lgbm	PAC_MCP	0.907	0.015	0.000	0.099	0.000	0.000

task	data_id	name	model	method	cov_novel	pi	pi_p	size_novel	bad	fallback
regression	416	yprop_4_1	lgbm	SCP	0.899	0.015	0.000	0.094	0.000	0.000
regression	416	yprop_4_1	logreg	DCP	0.899	0.015	0.000	0.098	0.000	0.000
regression	416	yprop_4_1	logreg	MCP	0.898	0.015	0.000	0.098	0.000	0.000
regression	416	yprop_4_1	logreg	PAC_MCP	0.907	0.015	0.000	0.103	0.000	0.000
regression	416	yprop_4_1	logreg	SCP	0.898	0.015	0.000	0.098	0.000	0.000
regression	416	yprop_4_1	rf	DCP	0.900	0.015	0.000	0.094	0.000	0.000
regression	416	yprop_4_1	rf	MCP	0.898	0.015	0.000	0.094	0.000	0.000
regression	416	yprop_4_1	rf	PAC_MCP	0.907	0.015	0.000	0.099	0.000	0.000
regression	416	yprop_4_1	rf	SCP	0.899	0.015	0.000	0.094	0.000	0.000

R4 Pre-specified tests P1, P2, P5

set	learner	n	rho_gstar	p_gstar_one_sided	rho_gpi	rho_diff_ci	share_above_bound	share_above_ci	median_ratio_obs_to_gstar	violations
prospective	rf	20	0.1203	0.3067	-0.2527	[-0.05904617713853144, 0.8260694588898585]	1.0000	[0.8315665290169147, 1.0]	-0.0292	[]
prospective	lgbm	20	0.3925	0.0435	0.0399	[-0.14753413510406319, 0.850289214334953]	1.0000	[0.8315665290169147, 1.0]	0.7866	[]
prospective	logreg	20	-0.0376	0.5625	0.0293	[-0.6712363919672428, 0.575768488524364]	0.9500	[0.7512672372279723, 0.9987349105020502]	-0.0431	[{'name': 'black_friday', 'pi': 0.6311, 'pi_p': 0.0002, 'g_star': -0.0, 'gap_scp

R5 Pre-specified test P3

learner	method	n	median_gap	ci_lo	ci_hi	worst	noninferior	median_bad_split_rate	mean_bad_split_rate	fallback_rate
rf	SCP	20	-0.0000	-0.0017	0.0098	-0.2754	True	0.0000	0.1050	0.0000
rf	MCP	20	0.0002	-0.0009	0.0024	-0.0069	True	0.0000	0.0425	0.0375
rf	PAC_MCP	20	0.0089	0.0062	0.0141	0.0035	True	0.0000	0.0050	0.0500
rf	DCP	19	0.0011	-0.0001	0.0046	-0.0008	True	0.0000	0.0184	0.0000
lgbm	SCP	20	-0.0022	-0.0099	0.0005	-0.3332	True	0.0000	0.1300	0.0000
lgbm	MCP	20	-0.0001	-0.0022	0.0004	-0.0092	True	0.0000	0.0450	0.0375
lgbm	PAC_MCP	20	0.0078	0.0060	0.0145	0.0025	True	0.0000	0.0075	0.0500
lgbm	DCP	19	0.0010	0.0000	0.0024	-0.0032	True	0.0000	0.0289	0.0000
logreg	SCP	20	-0.0006	-0.0040	0.0015	-0.0432	True	0.0000	0.1050	0.0000
logreg	MCP	20	0.0001	-0.0010	0.0016	-0.0052	True	0.0000	0.0475	0.0375
logreg	PAC_MCP	20	0.0094	0.0063	0.0152	0.0041	True	0.0000	0.0050	0.0500
logreg	DCP	19	0.0016	0.0001	0.0024	-0.0014	True	0.0000	0.0211	0.0000

R6 Pre-specified test P4

learner	n	mcp_bad_mean	pac_bad_mean	p_pac_lower
rf	20	0.0425	0.0050	0.0312
lgbm	20	0.0450	0.0075	0.0156
logreg	20	0.0475	0.0050	0.0312

R7 Normalized-residual score on fresh regression data

learner	method	n	median_gap	worst
lgbm	MCP	12	0.0000	-0.0163
lgbm	PAC_MCP	12	0.0078	0.0036
lgbm	SCP	12	0.0026	-0.0219
logreg	MCP	12	0.0011	-0.0048
logreg	PAC_MCP	12	0.0087	0.0022
logreg	SCP	12	0.0103	-0.0004
rf	MCP	12	0.0013	-0.0042
rf	PAC_MCP	12	0.0088	0.0035
rf	SCP	12	0.0070	-0.0243

Holm-adjusted primary p-values: {'P1_rf': 0.3067088322653912, 'P4_rf': 0.0625}.

R8 Near-copies: median gap on truly new rows (pp/100) by noise level

task	model	sigma	DCP	MCP_exact	MCP_oracle	SCP
classification	lgbm	0.0000	-0.0001	-0.0012	-0.0012	-0.0553
classification	lgbm	0.0010	-0.0008	-0.0323	-0.0008	-0.0403
classification	lgbm	0.0100	-0.0025	-0.0275	-0.0004	-0.0340
classification	lgbm	0.1000	-0.0004	-0.0224	-0.0004	-0.0228
classification	rf	0.0000	0.0048	0.0032	0.0032	-0.0581
classification	rf	0.0010	0.0020	-0.0411	-0.0008	-0.0430
classification	rf	0.0100	0.0011	-0.0346	-0.0002	-0.0364
classification	rf	0.1000	0.0050	-0.0241	0.0009	-0.0271
regression	lgbm	0.0000	0.0023	0.0032	0.0032	-0.0410
regression	lgbm	0.0010	0.0017	-0.0311	0.0025	-0.0311
regression	lgbm	0.0100	0.0007	-0.0267	0.0028	-0.0267
regression	lgbm	0.1000	0.0058	-0.0042	0.0019	-0.0042
regression	rf	0.0000	0.0066	0.0055	0.0055	-0.0519
regression	rf	0.0010	0.0025	-0.0411	0.0048	-0.0411
regression	rf	0.0100	0.0012	-0.0328	0.0053	-0.0328
regression	rf	0.1000	0.0050	-0.0081	0.0053	-0.0081

R9 Retrospective test of Eq. (3) (non-blind)

set	learner	n	rho_gstar	p_gstar_one_sided	rho_gpi	rho_diff_ci	share_above_bound	share_above_ci	median_ratio_obs_to_gstar
retrospective (v1 suites+external)	rf	41	0.7192	0.0000	0.5098	[-0.04590769219299606, 0.5003938114773784]	0.9756	[0.8714459796158537, 0.9993826831067135]	0.8418
retrospective (v1 suites+external)	lgbm	41	0.4915	0.0006	0.4688	[-0.204124397826055, 0.28194578049846963]	0.9512	[0.8346666465821697, 0.9940368817589572]	0.6936
retrospective (v1 suites+external)	mlp	41	0.6192	0.0000	0.5028	[-0.20576108574476273, 0.4607987817384518]	0.9756	[0.8714459796158537, 0.9993826831067135]	0.4437
retrospective (v1 suites+external)	logreg	41	0.4523	0.0015	0.6375	[-0.407650899295903, -0.004888652230568336]	0.9512	[0.8346666465821697, 0.9940368817589572]	0.4655
retrospective (v1 suites+external)	knn	41	0.1150	0.2370	0.2183	[-0.4126709261753157, 0.20357144249301057]	0.9756	[0.8714459796158537, 0.9993826831067135]	-0.2070

R10 Reinforcement protocol and deviations

Freeze point: the git commit that first adds this file. No reinforcement experiment (U2–U4) had been run, and no prospective dataset had been downloaded, when it was written. Everything in the v1 protocol stays as reported; this document adds to it.

1. Prospective validation set (U4)

- **Pool:** all datasets in OpenML studies 271 (AMLB classification), 269 (AMLB regression), 334, 335, 336, 337 (Grinsztajn et al. 2022 suites).
- **Exclusions (applied in order, recorded):**
  1. OpenML data id already used in v1 (103 suite + 5 external datasets).
  2. Dataset *name* (case-insensitive, ignoring version suffixes) already used in v1, to avoid re-entering the same data under another id.
  3. More than 2,000 features, or more than 5,000,000 rows (download feasibility). Image datasets are excluded.
  4. The download or parse fails (recorded).
- **Subsampling:** to at most 30,000 rows; whole record groups, except when the largest group exceeds 6,000 rows (20% of the cap), in which case uniform row subsampling is used (the rule that was forced on KDD Cup'99 in v1, now stated in advance). Seed 2026.
- **Inclusion:** label-free predicted exposure $\hat{\pi} \geq 0.01$ (the v1 DUP definition), computed after subsampling. Inclusion uses **no labels and no models**.
- **Learners:** RF (200 trees), LightGBM (300 trees), LR/ridge, as in v1. **Scores:** LAC (classification), |res| (regression), plus the normalized residual (regression; v1 definition). **α = 0.1. Seeds 0–19. Protocol R** (60/20/20 random split).
- **Methods:** SCP, MCP (exact-key seen/novel), PAC-MCP (δ = 0.1), DCP (six-bin distance-Mondrian, v1 definition), and for regression SCP and MCP with the normalized score.

2. Pre-specified quantities

- **π_P (label purity–adjusted exposure):** for each split, the share of test rows that (i) have ≥ 1 training row with an identical feature key and (ii) whose training copies **all** carry exactly the test row's label (classification) or target value (regression). Averaged over seeds.
- **Purity-adjusted bound:** $g^* = -\alpha\ \pi_P/(1-\pi_P)$ ($-\alpha$ if $\pi_P \geq 1-\alpha$).
- **Exposure-only bound:** $g_\pi = -\alpha\ \pi/(1-\pi)$.
- **Observed gap:** mean over seeds of (novel-row coverage − 0.9) under SCP.

3. Pre-specified tests (prospective set; unit = dataset)

- **P1 (discrimination).** Spearman ρ(g*, observed SCP gap) for RF with LAC or |res| (primary learner, pooled over tasks): one-sided test of ρ > 0. Secondary: the same for LightGBM and LR.
- **P2 (bound).** For RF, the share of datasets whose observed gap lies above $g^* - 0.02$ (the bound holds up to noise). Hypothesis: ≥ 80%. Reported with an exact binomial 95% CI.
- **P3 (remedy).** Median MCP-R and PAC-MCP novel gaps: bootstrap 95% CI lower bound > −0.01 (RF, LightGBM, LR).
- **P4 (training-conditional).** Per dataset, the fraction of seeds with novel coverage below 0.87, for MCP vs PAC-MCP: one-sided paired Wilcoxon test (PAC-MCP lower).
- **P5 (purity vs exposure).** Spearman ρ of g* vs ρ of g_π with the observed RF gap; the difference is reported with a dataset-bootstrap 95% CI (10,000 resamples).
- **Multiplicity:** Holm across the primary tests {P1 (RF), P3 (RF, MCP), P4}. Everything else is secondary.
- **Retrospective analysis (non-blind, labelled as such):** the same P1/P2/P5 quantities on the v1 suites (DUP), the external set and the injection experiment.

4. Near-duplicate injection (U3)

- **Datasets:** the 37 duplicate-free datasets of the v1 injection experiment (≥ 1,000 rows, randomly capped at 10,000 rows, v1 seed rule).
- **Design:** append jittered copies of a random 50% of rows (ρ = 0.5). Each numeric feature gets jitter $N(0, (\sigma \cdot sd_j)^2)$, $\sigma \in \{0, 10^{-3}, 10^{-2}, 10^{-1}\}$, where sd_j is the original feature's standard deviation. Categorical features and labels are copied unchanged. Seeds 0–2; RF and LightGBM; LAC or |res|; α = 0.1; protocol R.
- **Ground-truth novelty:** a test row is *truly new* if no training row descends from the same original record (source id tracked through the injection).
- **Methods:** SCP; MCP-exact (exact-key taxonomy); DCP (distance-Mondrian, v1 definition); MCP-oracle (taxonomy = record identity; infeasible in practice, used as a reference).
- **Endpoints (descriptive):** N1, the median SCP gap on truly new rows versus σ. N2, the median gap of MCP-exact, DCP and MCP-oracle versus σ. N3, the share of injected copies detected as "seen" by the exact key, versus σ.

5. Reporting

All prospective datasets are reported. Failed downloads, failed fits and infeasible splits are logged and reported. No seed or dataset is selected by outcome.

Amendment 1 (2026-10-21 ~21:05 UTC, before any prospective dataset was downloaded or run)

Reason. Since 20:50 UTC every connection from this environment to openml.org (www, api and test hosts) is closed by the network proxy (`ws_closed_mid_exchange`). The UCI repository and PyPI remain reachable.

Contingency rule. * If OpenML is still unreachable at 22:50 UTC (two hours after the first failure), the prospective pool of §1 is replaced by **every dataset listed by the UCI Machine Learning Repository API** (`ucimlrepo.list_available_datasets()`; the snapshot is saved with the selection file). They are loaded with `fetch_ucirepo(id)`. If OpenML becomes reachable before then, §1 applies unchanged. * UCI-specific eligibility, applied in this order and recorded: (i) exactly one target column that is non-missing; (ii) task taken from the UCI metadata (`Classification` → classification; else `Regression` → regression; otherwise excluded); (iii) 300 ≤ rows ≤ 5,000,000 and ≤ 2,000 features. * **Overlap with v1 (content-based, conservative).** Names differ between UCI and OpenML (e.g. "Breast Cancer Wisconsin (Original)" is OpenML *breast-w*), so a UCI dataset is excluded if its number of rows lies within ±2% of a v1 dataset whose number of features differs by at most 1, or if its normalized name matches a v1 name. This may exclude some unrelated datasets, which is acceptable. * Everything else (subsampling rule, inclusion $\hat{\pi} \geq 0.01$, learners, seeds, methods, tests P1–P5) is unchanged.

Deviation note on Amendment 1 (2026-10-29)

The session was suspended between 2026-10-21 ~21:15 UTC and 2026-10-29, so the 22:50 UTC check was never performed. When the prospective stage was finally run (2026-10-29 13:08 UTC), OpenML was reachable again, and **no dataset had been downloaded from either source**. The primary pool (§1, OpenML) was therefore used unchanged, and the UCI contingency was not triggered. The UCI script (`code/experiments/prospective_selection_uci.py`) is kept but was not run.

Deviation 2: selection details (2026-10-29, before any prospective outcome was computed)

1. **Parsing bug fixed.** The first selection attempt crashed because some OpenML “qualities” are lists, not numbers. The parser now skips non-numeric qualities (they are not used). No dataset was evaluated before the fix.
2. **Same data under two ids.** The selection (24 of 163 included) contained four data sets twice: wine_quality (287, 44136), yprop_4_1 (416, 45032), Mercedes_Benz_Greener_Manufacturing (42570, 44061. and nyc-taxi-green-dec-2016 (44065, 44143); the Grinsztajn suites re-host AMLB data. §1 de-duplicated only against v1. To avoid counting a data set twice, one copy per normalized name is kept, the one with the lowest id. The rule uses names only (no labels, no outcomes). This leaves **20** data sets.
3. **Transient download failures.** Seven downloads failed. Two are genuine parse failures (sparse ARFF, QSAR-TID-11 and QSAR-TID-10980) and stay excluded. Five were server-side (HTTP 503, “not found” while the API returned 503, or checksum errors, which were probably caused by two parallel download helpers writing the same file). These five were retried: Satellite (40900) succeeded at once. The others are retried every 2 minutes for at most 60 minutes, a cut-off set before retrying. Any data set that is still unavailable at the cut-off stays excluded as “download failed”, as §1 prescribes. Data sets added by a successful retry go through the same label-free inclusion rule.
4. **Early start.** To save time, the prospective runs for the 20 data sets fixed by points 1–2 were started while the retries were pending. The code and configuration are identical to `run_all.py --stage prospective`.
5. **Outcome of the retries (cut-off reached 15:20 UTC).** Satellite (40900) and volkert (41166) were downloaded; both have predicted exposure below 0.01 and are excluded by the inclusion rule. albert (41147) still returned HTTP 503 at the cut-off. Yolanda (42705) and Higgs (42769) failed OpenML’s own md5 check on every attempt. A last attempt with the local download cache cleared fetched both files afresh and they still failed, so the mismatch is on the server side. These three stay excluded as “download failed”. A selection re-run with the de-duplication rule (`results/logs/prospective_selection_rerun1.log`) reproduced the same **20** data sets.